

International Baccalaureate Physics Extended Essay

How do relevant parameters affect the normal-mode frequencies and thereby influence energy transport and localization in a chain of magnetic coupled pendulums?

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1 Introduction

1.1 Research motivation

A chain of magnetically coupled pendulums provides a macroscopic analogue of more complex systems in which interacting particles exchange and transport vibrational energy. The pendulums represent discrete sites, gravity provides an on site potential, and magnetic repulsion represents the nonlinear interaction between neighbouring sites. Similar mathematical structures arise in models of crystal lattices, phonon transport, mechanical metamaterials, and other discrete nonlinear media. Unlike microscopic systems, the displacement of every pendulum can be observed directly, while the coupling strength, lattice spacing, damping, and initial excitation can be controlled experimentally. The model therefore allows processes such as dispersive wave propagation, modal energy exchange, recurrence, and energy localization to be studied in an accessible and measurable form. Understanding how these behaviours depend on the normal mode frequencies may provide insight into the mechanisms that govern vibrational energy transport in more complex discrete systems.

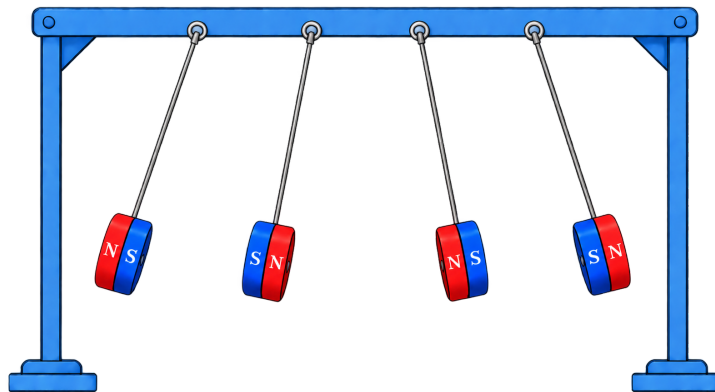


Figure 1: Cartoon of the investigated system. Identical pendulums carry like-oriented cylindrical magnets, so neighbouring bobs repel without contact. One pendulum is released from an angle θ_0 ; the resulting motion is recorded for every site.

1.2 Research background

Nonlinearly coupled oscillator lattices have been used as reduced models of collective dynamics across a wide range of physical scales. At the microscopic scale, trapped ions form ordered structures in which Coulomb repulsion provides interparticle coupling and an external electromagnetic field maintains confinement [1]. At the mesoscopic scale, granular chains represent discrete elastic media whose particles interact through nonlinear Hertzian contact forces [2, 3]. At the macroscopic scale, sonic and phononic crystals use periodic variations in material properties to control acoustic dispersion, harmonic generation, and band gap formation [4, 5]. Although the physical interactions differ, these systems share two essential features. Their discrete periodic

structure produces a nontrivial dispersion relation, while their nonlinear interactions allow energy to be transferred between frequencies and wave numbers. Wave propagation is therefore determined by the combined effects of dispersion, discreteness, and nonlinearity.

Magnetic repulsion provides a contactless and experimentally adjustable realization of a nonlinear oscillator lattice. The interaction is strongly dependent on separation and can often be approximated by an inverse power law over a restricted experimental range. Chains of repelling magnets have been used to generate solitary waves whose velocity and spatial width depend on excitation amplitude [6]. In harmonically driven magnetic lattices, nonlinear coupling produces higher harmonics, while the dispersion relation determines whether these harmonics propagate, undergo spatial beating, or become evanescent [7, 8]. When the magnetic elements are attached to pendulums, gravity introduces an additional on site restoring potential, producing a nonzero lower cutoff frequency and modifying the conditions for wave propagation [8, 9]. Magnetic interactions can also be modified to produce effective nonreciprocal coupling. For example, a moving conductor can alter the fields generated by magnetic dipoles and produce asymmetric energy exchange [10]. Magnetic oscillator systems therefore provide a versatile platform for studying both conventional reciprocal lattices and more general forms of coupled dynamics.

A major direction in nonlinear lattice research concerns the formation of spatially localized excitations. Discrete breathers are time periodic oscillations whose energy remains concentrated around a small number of lattice sites. Their existence is associated with the combination of nonlinearity and discreteness, which can prevent a localized oscillation from resonantly radiating its energy into extended linear modes [11]. Moving breathers have been observed in a ring of magnetic pendulums, where sufficiently strong localized impulses formed compact propagating wave packets [9]. Related localization has been observed in diatomic granular chains, where modulational instability near a spectral band edge produced long lived discrete breathers within a frequency gap [3]. Magnetic particle chains have also supported amplitude dependent solitary waves [6], while realistic models of muscovite mica have predicted highly localized supersonic kinks associated with energy transport along preferred crystallographic directions [12]. These studies distinguish several forms of localized motion, including solitary waves, kinks, and breathers, whose stability depends on the force law, dispersion relation, on site potential, and excitation amplitude.

Energy localization can also occur in normal mode space without remaining localized at a particular position. The Fermi-Pasta-Ulam problem showed that weak nonlinearity does not necessarily produce rapid equipartition among normal modes. Instead, energy initially placed in a small number of modes may return close to its initial distribution. This behaviour is associated with recurrence and weak resonant coupling between modes. The concept of a q breather formalizes this effect as a nonlinear periodic orbit that remains exponentially localized in normal mode space [13]. More general q torus solutions have been used to explain long term confinement of energy within a limited group of low frequency modes [14]. The importance

of modal frequency separation has also been demonstrated in magnetically coupled pendulums. Controlled variation of the coupling between two pendulums changes the separation of the in phase and out of phase eigenfrequencies and determines the rate and completeness of coherent energy exchange [15]. These results show that recurrence and spatial localization are physically distinct phenomena, even though both originate from the interaction between the linear frequency spectrum and nonlinear mode coupling.

Existing research therefore identifies a common sequence connecting lattice parameters to dynamics. The interaction law and on site potential determine the normal mode frequencies and dispersion relation. The resulting mode spacing and group velocities govern linear propagation, while nonlinear frequency shifts and resonances determine whether energy spreads, returns to a restricted set of modes, becomes irregular, or remains localized in real space. However, these components have generally been investigated separately in long driven chains, periodic rings, idealized conservative models, or systems containing only two oscillators. The systematic connection between experimentally adjustable parameters, the complete eigenfrequency spectrum of a finite magnetic pendulum chain, and the resulting transport and localization of energy remains insufficiently established.

1.3 Research question and hypothesis

Research question: How do relevant parameters affect the normal-mode frequencies and thereby influence energy transport and localization in a chain of magnetic coupled pendulums?

The normal mode frequencies are hypothesized to provide the principal connection between the physical parameters of the chain and its energy dynamics. Increasing the magnetic coupling strength, either by increasing magnetization or reducing the equilibrium separation, should increase the effective coupling stiffness. This should widen the frequency spectrum and produce the largest increase in modes containing substantial relative motion between neighbouring pendulums. The nearly in phase mode should be less sensitive to magnetic coupling because adjacent magnetic separations change only slightly in this mode. Its frequency should remain primarily determined by the gravitational restoring torque.

Increasing the number of pendulums should increase the number and density of allowed modes. Changes in pendulum length should modify the gravitational frequency scale, while changes in release amplitude should introduce nonlinear frequency shifts through both the gravitational torque and the magnetic interaction. The resulting changes in mode spacing and dispersion should determine which modes can exchange energy efficiently. A spectrum containing closely spaced or near resonant modes is expected to promote modal energy transfer and dispersive spreading. Strong nonlinear frequency shifts may instead reduce resonance with extended modes and favour recurrence or spatial localization. Damping is expected to reduce the duration and visibility of all three behaviours without determining the normal mode frequencies to first order.

1.4 Scope and approach

The research question is addressed in two stages. In the first stage (Sections 3–7), I measured the normal-mode frequencies of chains of two to five pendulums while varying spacing, magnetic moment, pendulum length and release angle, and compared them with a Lagrangian model built from independently measured parameters. In the second stage (Sections 8–9), this model is used as a numerical tool to examine how the mode spectrum governs energy transport and localization. The simulation results are therefore predictions of the tested model, and their range of validity is evaluated in Section 11.

2 Variables

The experiment is a multi-factor investigation. Only one factor was intentionally changed within each trial; all other quantities were held at constant.

Table 1: Operational definition and control of variables.

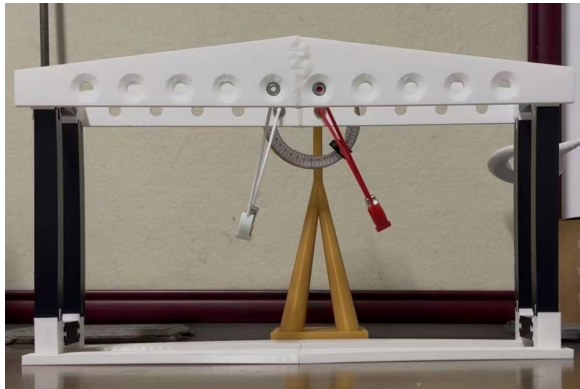
Type	Variable	measurement/range	Control method
Independent	Number of pendulums N	Number of active, equally spaced magnetic pendulums; $N = 2\text{--}5$.	Unused supports were kept outside the interaction region; the same magnet and arm design was used. N changes the number of degrees of freedom and standing waves.
Independent	Release angle θ_0	Initial angular displacement read from the protractor and verified from the Tracker video analysis. Varies from $\theta = 25^\circ$ to $\theta = 60^\circ$	Only the end pendulum was displaced; the remaining pendulums began at equilibrium positions.
Independent	Pivot spacing s	Centre-to-centre horizontal distance between adjacent pivots, set by the sliding PETG top brackets. Varies from $s = 0.04\text{ m}$ to $s = 0.16\text{ m}$	All spacings within one run were equal.

Type	Variable	measurement/range	Control method
Independent	Pendulum length ℓ	Pivot to the reference point used in the coordinate model; Varies from $\ell = 0.089 \text{ m}$ to $\ell = 0.162 \text{ m}$	Identical arms were fitted within a trial
Independent	Magnetic moment m / magnetization $\mathcal{M}$	Deduced from an axial field trial. The fitted values were $\mathcal{M} = (1.12 \pm 0.04) \times 10^6 \text{ A m}^{-1}$. Varies from $m = 4.00 \text{ A m}^2$ to $m = 12.00 \text{ A m}^2$.	Magnet orientation was kept repulsive.
Dependent	Eigenfrequencies f_m	Peak frequencies in the FFT of $\theta_n(t)$; $f_m = \omega_m/(2\pi)$. Phase and amplitude across sites were used to distinguish modes from harmonics.	Common frame rate and observation time were used within a trial.
Secondary response	Modal amplitudes, phase, MLE, IPR	FFT magnitude and phase; maximum Lyapunov exponent in simulation; real-space inverse participation ratio in the nonlinear extension.	These diagnose which modes are excited and whether motion is regular, chaotic or localized.
Controlled	Magnet geometry	Cylinders of radius $R = 15 \text{ mm}$ and height $H = 10 \text{ mm}$ in the reference model.	The same batch and orientation were used; damaged or chipped magnets were excluded.
Controlled	Oscillating mass and inertia	Magnet mass approximately $M = 50.0 \text{ g}$ plus arm and basket contributions.	Components were not exchanged within a trial.

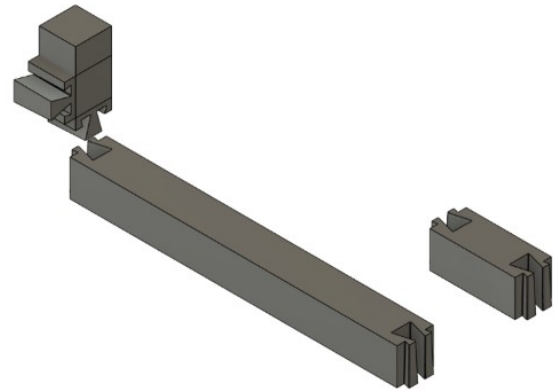
3 Experimental setup

3.1 Complete apparatus

The apparatus consisted of a rigid top rail, adjustable non-ferromagnetic brackets, bearing-mounted pendulum arms, cylindrical permanent magnets housed in baskets, a release-angle protractor, and a camera positioned normal to the plane of motion. Non-ferromagnetic PETG components were used near the magnets so that the support did not create an uncontrolled magnetic restoring force.



(a) Physical apparatus used in the experiment.



(b) CAD assembly used to define the geometry.

Figure 2: Experimental setup. The camera viewed the pendulum plane approximately normally. Sliding top brackets set the equal pivot spacing s ; the arm geometry set ℓ .

Top bracket and connector. The upper PETG bracket suspended each bearing and slid along the support to set s . Its connector fixed the bearing height, so all pivots lay on one horizontal line. This alignment is important because a vertical offset produces unequal equilibrium separations and breaks the assumed reflection symmetry.

Pendulum arm, bearings and basket. The arm fixed the pivot-to-magnet geometry. Two bearings reduced pivot friction and constrained rotation to one plane. The lower basket held the magnet mechanically and prevented it from flipping into the attractive orientation.

Protractor. A protractor attached to the end support set the nominal release angle. The angle from the video was used as the initial condition in the simulation.

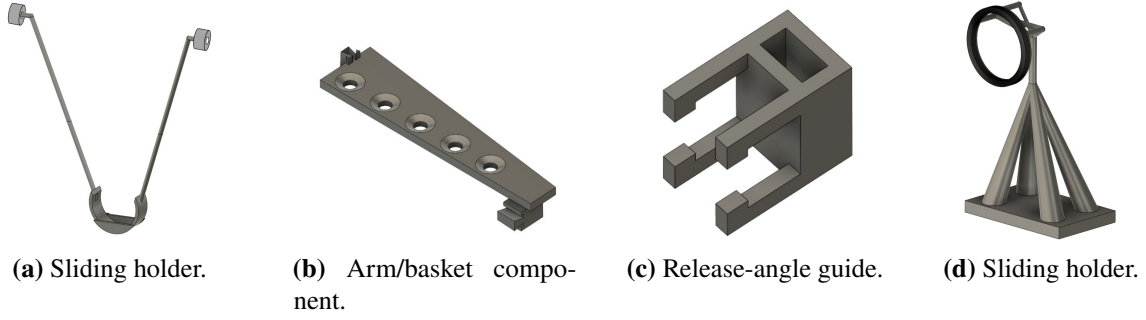


Figure 3: Principal components.

3.2 Magnet characterization

The cylindrical magnet was modelled as uniformly magnetized along its axis. A uniform magnetization $\mathcal{M}$ is equivalent, on the curved surface, to a bound surface-current density

$$\mathbf{K}_b = \mathcal{M} \times \hat{n}, \quad (1)$$

so a thin slice of thickness dz' behaves as a circular current loop $dI = \mathcal{M} dz'$. For a source point

$$\mathbf{r}' = (R \cos \phi, R \sin \phi, z'), \quad d\ell' = R d\phi (-\sin \phi, \cos \phi, 0), \quad (2)$$

the field is obtained from the Biot–Savart integral

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_{-H/2}^{H/2} \int_0^{2\pi} \frac{\mathcal{M} d\ell' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} dz'. \quad (3)$$

On the symmetry axis this reduces to

$$B_z(z) = \frac{\mu_0 \mathcal{M}}{2} \left[\frac{z + H/2}{\sqrt{(z + H/2)^2 + R^2}} - \frac{z - H/2}{\sqrt{(z - H/2)^2 + R^2}} \right]. \quad (4)$$

Since the ruler reading x was not measured from the magnet centre, the axial distance in equation (4) was taken as $z = x + x_0$, where the offset $x_0 \approx 1.8$ cm.

Fitting equation (4) to the measured axial field gave

$$\mathcal{M} = (1.12 \pm 0.04) \times 10^6 \text{ A m}^{-1}. \quad (5)$$

The corresponding dipole moment is $m = \mathcal{M}V = \mathcal{M}\pi R^2 H \approx 7.89 \text{ A m}^2$ for the stated dimensions. The equivalent total surface current is $I_{\text{eq,mag}} = \mathcal{M}H$.

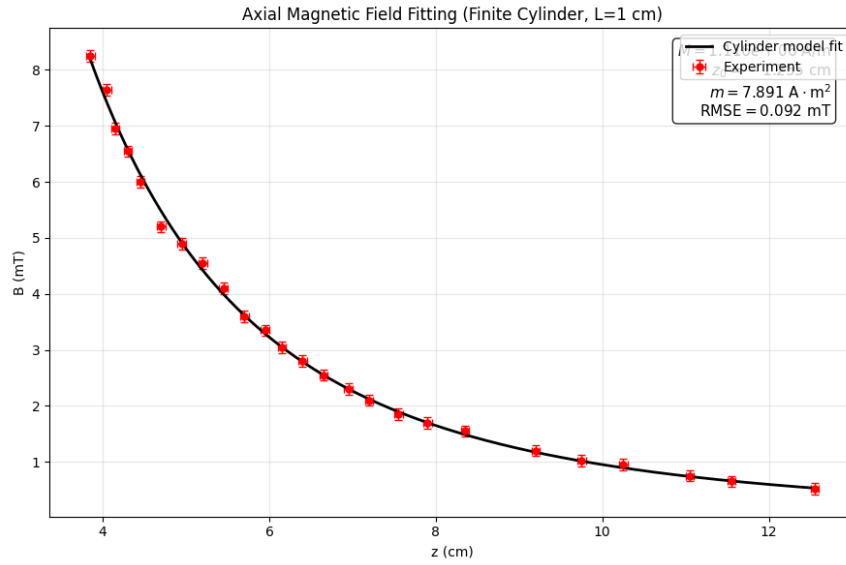


Figure 4: Measured on-axis field and fitted finite-cylinder model. $m = 7.891 \text{ A m}^2$ and an RMSE of approximately 0.092 mT.

3.3 Pendulum damping coefficient

An isolated pendulum was released at small amplitude and its peak envelope fitted to

$$\theta_{\text{env}}(t) = \theta_0 e^{-\lambda t}. \quad (6)$$

The fit yielded

$$\lambda = (0.0326 \pm 0.0009) \text{ s}^{-1}. \quad (7)$$

For $I_{\text{eq}}\ddot{\theta} + \gamma\dot{\theta} + K\theta = 0$ in the underdamped limit, $\lambda = \gamma/(2I_{\text{eq}})$, giving

$$\gamma = 2I_{\text{eq}}\lambda = (4.33 \pm 0.12) \times 10^{-5} \text{ kg m}^2 \text{ s}^{-1}. \quad (8)$$

The estimated Reynolds number was about 556 for $\rho = 1.2 \text{ kg m}^{-3}$, $\mu = 1.8 \times 10^{-5} \text{ Pa s}$, diameter 0.03 m, and characteristic speed $v \approx 2\pi f \ell \sin 20^\circ \approx 0.278 \text{ m s}^{-1}$. Thus the linear damping term is an empirical amplitude-range approximation.

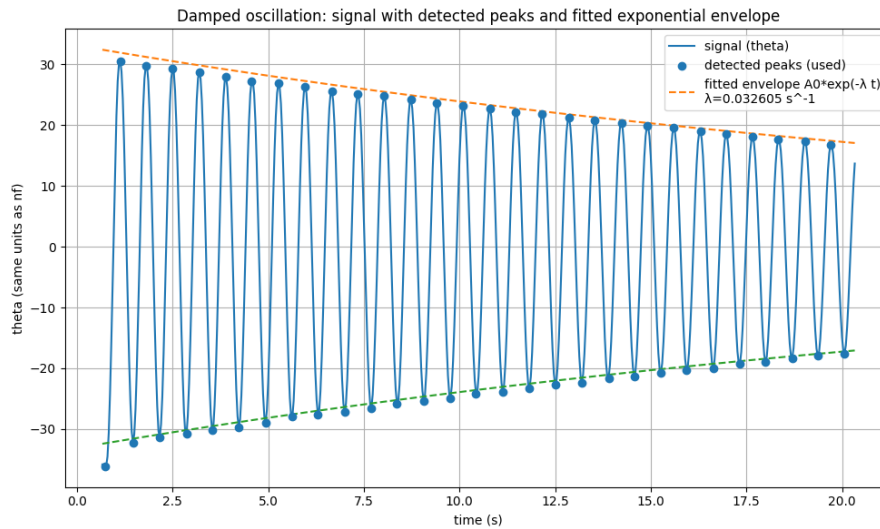


Figure 5: Independent decay measurement used to determine the linear damping coefficient.

4 Method

4.1 Experimental procedure

For each parameter setting, the following procedure was used.

1. Assemble N identical pendulums and set every neighbouring pivot separation to the selected s . Measure the pivot-to-reference length ℓ and verify that magnets have the same repulsive orientation.
2. Place the camera so its optical axis is normal to the pendulum plane. Include the protractor or another object of known length in the image for calibration. Lock focus, exposure and camera position.
3. Allow every pendulum to settle. Displace the chosen end pendulum to θ_0 without touching its neighbours. Start recording before release so the equilibrium and initial condition are visible.
4. Release without an intentional push and record until several energy-transfer cycles have occurred or amplitudes have decayed near the tracking-noise floor. Repeat the run at least 3 times.
5. Change only the selected independent variable. Recheck alignment and equilibrium after every change of spacing, length, magnet stack or pendulum number.

4.2 Video tracking

Each pivot was set as a fixed origin in Tracker and the magnet/basket centre was tracked through the video. The coordinate axes and angular scale were calibrated using the apparatus geometry. Video analysis data has intervals of approximately 0.0333 s, corresponding to 30 frames s^{-1} .

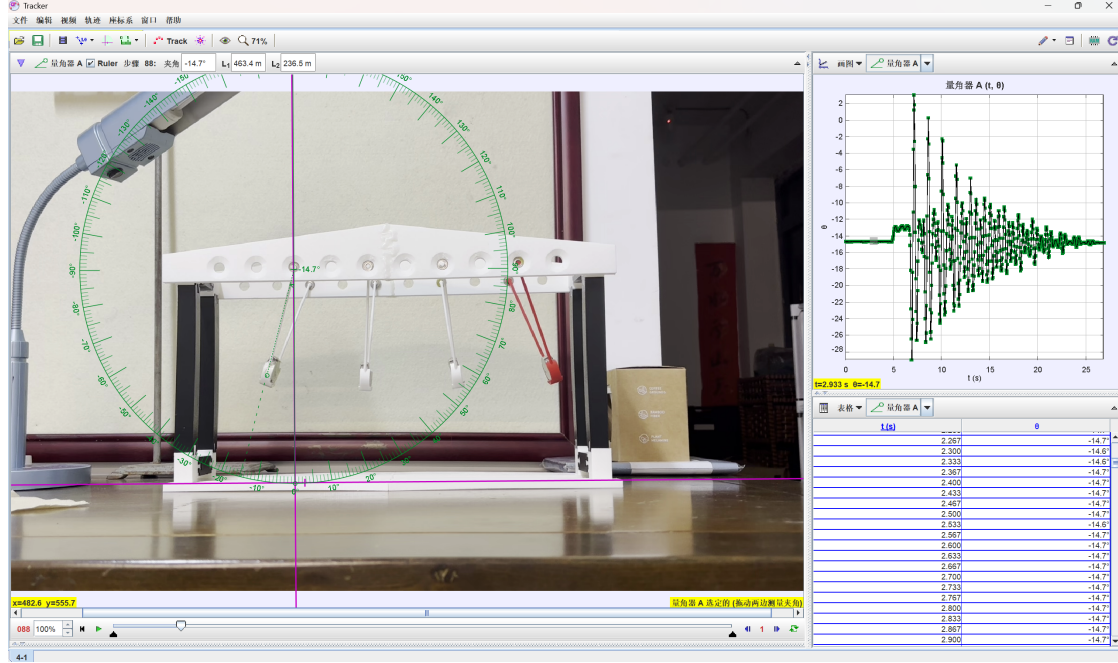


Figure 6: Tracker analysis screen from the experiment. The calibrated protractor and tracked bob position convert video frames into $\theta_n(t)$ for each pendulum.

The equilibrium offset of each trace was removed:

$$x_n(t_j) = \theta_n(t_j) - \overline{\theta_n}. \quad (9)$$

4.3 FFT and eigenfrequency extraction

For N_t equally spaced samples with spacing Δt , the discrete Fourier transform was calculated as

$$X_n(f_k) = \sum_{j=0}^{N_t-1} x_n(t_j) \exp\left(-2\pi i \frac{jk}{N_t}\right), \quad f_k = \frac{k}{N_t \Delta t}. \quad (10)$$

Only positive frequencies were retained. A genuine collective mode was identified when a peak appeared at a common frequency in more than one pendulum and its complex phases were compatible with a mode shape. Peaks close to integer multiples of a stronger fundamental were marked as possible nonlinear harmonics.

The nominal FFT-bin spacing was

$$\Delta f_{\text{bin}} = \frac{1}{T_{\text{record}}}. \quad (11)$$

4.4 Theory–experiment comparison

The nonlinear equations in section 6 were integrated by fourth-order Runge–Kutta using measured initial angles and zero initial velocities. The same record duration and sampling interval were then applied to the simulated trace before the FFT. The comparison quantities were

$$\Delta f_m = f_{m,\text{exp}} - f_{m,\text{model}}, \quad (12)$$

$$\varepsilon_m = \frac{|\Delta f_m|}{f_{m,\text{model}}} \times 100\%, \quad (13)$$

$$\text{RMSE}_f = \sqrt{\frac{1}{M} \sum_{m=1}^M (\Delta f_m)^2}. \quad (14)$$

5 Preliminary investigation

The preliminary experiments were conducted to qualitatively investigate the influence of different parameters on the system's behaviors.

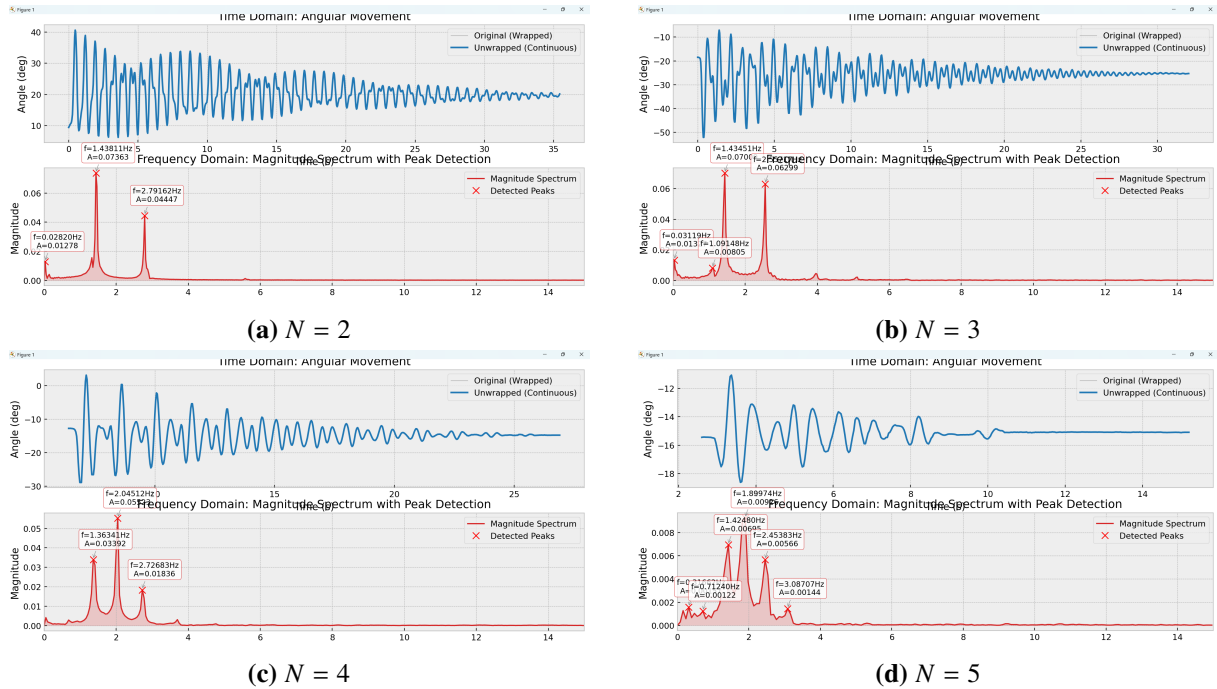


Figure 7: Preliminary FFT records as pendulum number was increased. More degrees of freedom produced more resolvable common peaks, supporting the presence of several normal modes.

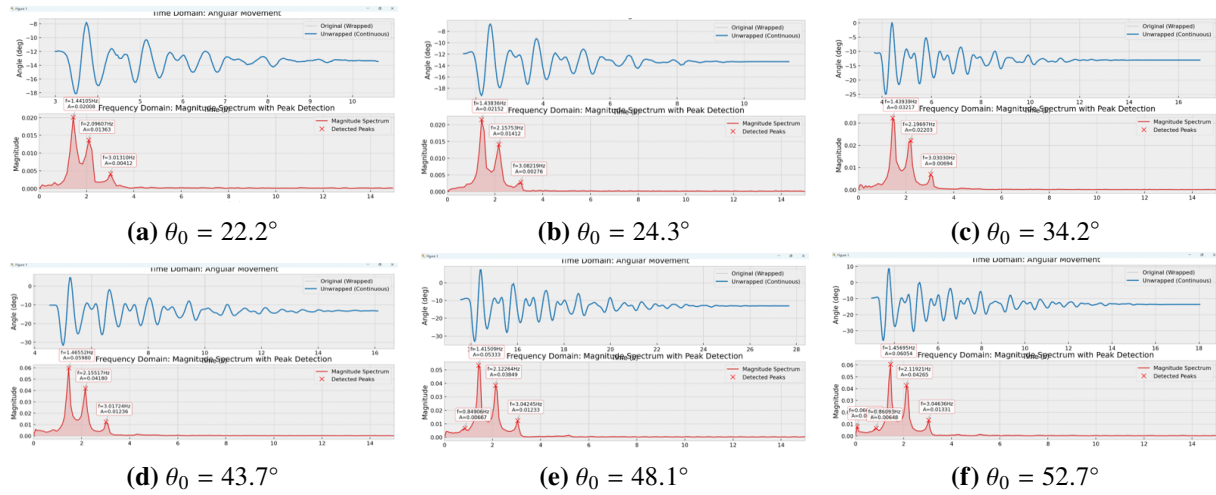


Figure 8: Preliminary displacement and FFT records obtained at different initial release angles. The frequency peaks shift and their relative amplitudes change as the oscillation amplitude increases, indicating an amplitude-dependent response of the magnetically coupled system.

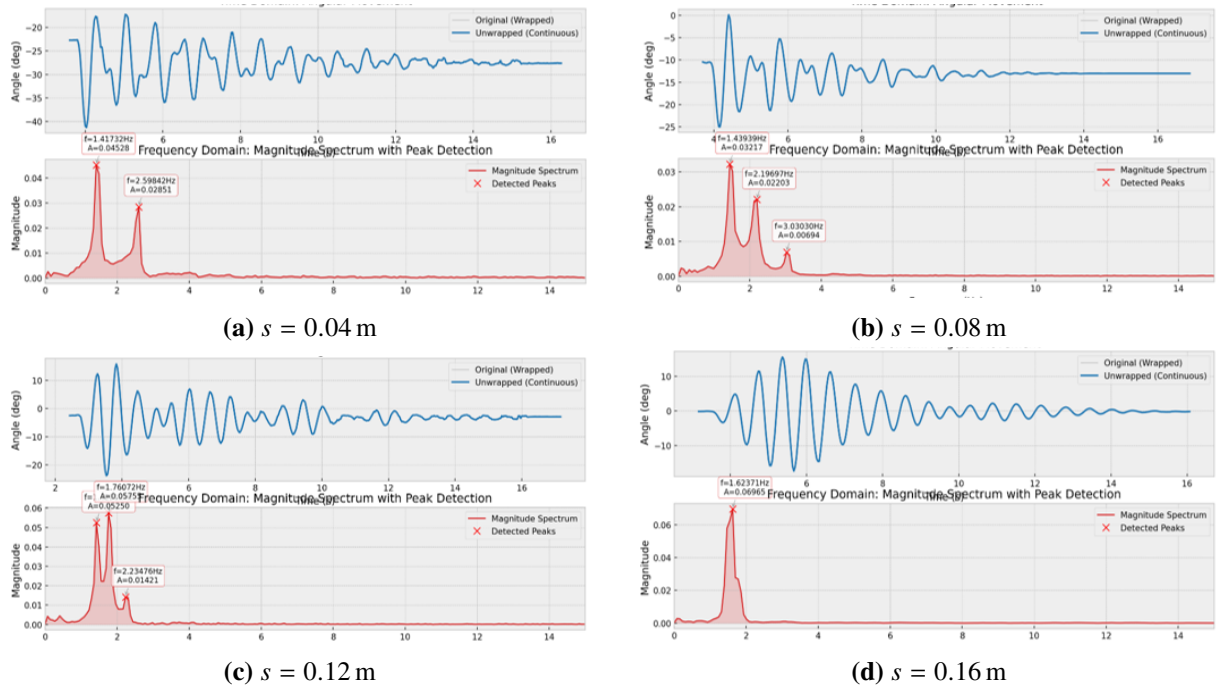


Figure 9: Preliminary displacement and FFT records obtained at different equilibrium spacings. Increasing the separation weakens the magnetic coupling and changes the separation and relative prominence of the dominant frequency peaks.

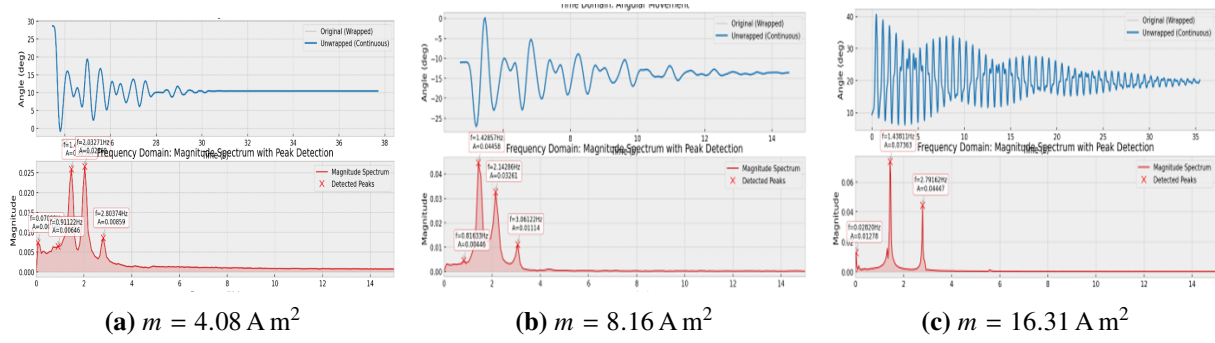


Figure 10: Preliminary displacement and FFT records obtained using different total magnetic moments. Increasing the magnetic moment strengthens the coupling between adjacent pendulums and produces a larger change in the magnetically influenced frequency components.

Four qualitative results guided the final method:

1. Increasing N increased the number of spectral peaks, approximately as expected for N degrees of freedom, although not every mode was equally excited by an end release.
2. Increasing θ_0 shifted the dominant frequencies, showing that a purely small-angle analysis would not describe all runs.
3. Increasing s changed the separation of the low and high branches; the higher, out-of-phase peak changed much more strongly.
4. Increasing magnetic moment mainly raised the second dominant frequency while leaving the lowest branch comparatively stable.

These observations led to the hypothesis in section 1.3: the nearly in-phase mode contains little relative neighbour displacement and is gravity dominated, while out-of-phase modes bend the magnetic “bonds” and are magnetically sensitive. The preliminary study also revealed that spectra alone are ambiguous at high amplitude. Consequently, later analysis uses both frequency and phase/mode shape.

6 Equations of motion

6.1 Assumptions and coordinates

The model used the following assumptions.

1. All magnets are identical, uniformly axially magnetized cylinders.
2. The support is rigid and all pivot axes are parallel; motion is planar.
3. Bearing and aerodynamic losses can be represented by one linear rotational damping coefficient γ over the measured amplitude range.

4. Each arm/basket/magnet assembly is rigid. The magnetic force follows the independently measured field/force law.

Let

$$\mathbf{q} = (\theta_1, \theta_2, \dots, \theta_N)^\top. \quad (15)$$

With horizontal pivot spacing s and pivot-to-magnet-centre lever arm approximated by ℓ , the magnet-centre position is

$$\mathbf{p}_i = ((i-1)s + \ell \sin \theta_i, -\ell \cos \theta_i, 0). \quad (16)$$

Therefore

$$r_{ij} = \left([(j-i)s + \ell(\sin \theta_j - \sin \theta_i)]^2 + [\ell(\cos \theta_i - \cos \theta_j)]^2 \right)^{1/2}. \quad (17)$$

6.2 Kinetic energy and equivalent inertia

The total moment of inertia is:

$$I_{\text{eq}} = M \left(\frac{R^2}{4} + \frac{H^2}{12} + (\ell + R)^2 \right) + \frac{1}{3} M_{\text{rod}} L_{\text{rod}}^2, \quad (18)$$

where the first bracket applies the parallel-axis theorem to a cylindrical magnet and the second term approximates the rod about one end. The total kinetic energy is

$$T = \frac{1}{2} I_{\text{eq}} \sum_{i=1}^N \dot{\theta}_i^2. \quad (19)$$

6.3 Gravitational and magnetic potential energy

If the magnet's centre of mass is a distance $d \approx \ell + R$ below the pivot,

$$V_g = \sum_{i=1}^N M g d (1 - \cos \theta_i). \quad (20)$$

The exact magnetic energy of magnet j in the field produced by magnet i is

$$V_{\text{mag},ij} = - \int_{V_j} \mathcal{M}_j \cdot \mathbf{B}_i(\mathbf{r}) dV, \quad (21)$$

and the full potential is

$$V(\mathbf{q}) = V_g + \sum_{j < i} V_{\text{mag},ij}. \quad (22)$$

6.4 Lagrange equation with Rayleigh dissipation

The Lagrangian and Rayleigh dissipation function are

$$\mathcal{L} = T - V, \quad \mathcal{R} = \frac{1}{2}\gamma \sum_{i=1}^N \dot{\theta}_i^2. \quad (23)$$

Applying

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_k} \right) - \frac{\partial \mathcal{L}}{\partial \theta_k} + \frac{\partial \mathcal{R}}{\partial \dot{\theta}_k} = 0 \quad (24)$$

gives the nonlinear coupled equations

$$I_{eq} \ddot{\theta}_k + \gamma \dot{\theta}_k + \frac{\partial V}{\partial \theta_k} = 0, \quad k = 1, \dots, N. \quad (25)$$

For the measured power-law, nearest-neighbour reduction and small geometric slope, the interior equation becomes

$$I_{eq} \ddot{\theta}_n + \gamma \dot{\theta}_n + Mgd \sin \theta_n + \beta \ell \left[(s + \ell(\theta_{n+1} - \theta_n))^{-\alpha} - (s + \ell(\theta_n - \theta_{n-1}))^{-\alpha} \right] = 0. \quad (26)$$

The exact numerical model evaluates the chosen magnetic energy and its derivatives.

6.5 Theory-Experiment comparison

The equations were integrated using measured parameters and initial conditions. For a representative two-pendulum calculation, the simulated motion alternated energy between pendulums and its FFT gave dominant peaks at approximately 1.433 Hz and 2.733 Hz. As shown in Fig.12, theoretical predictions agree well with the actual experimental data.

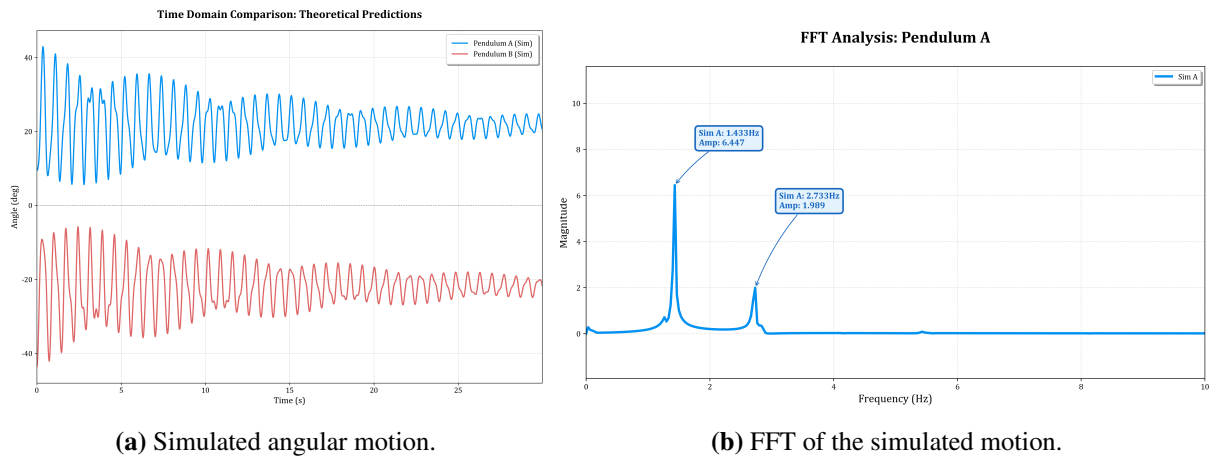


Figure 11: Representative Runge–Kutta solution of the nonlinear equations and its frequency spectrum.

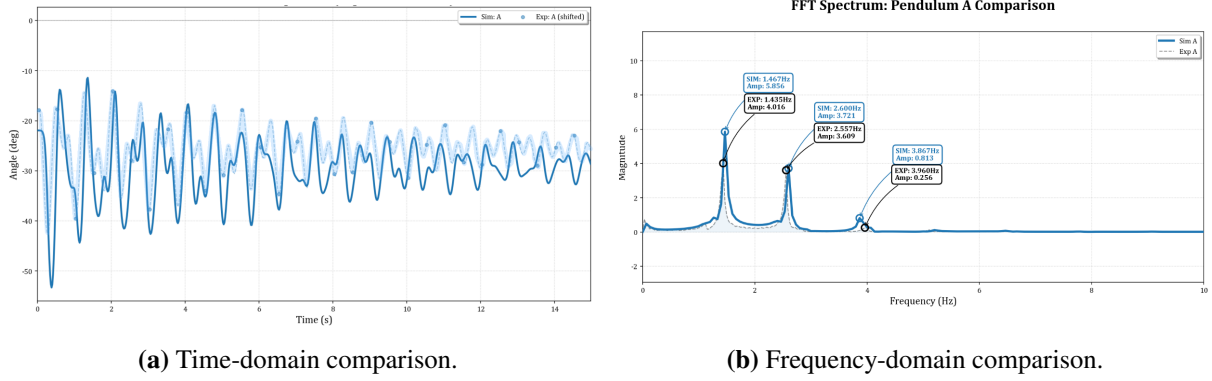


Figure 12: Theory–experiment comparison for a representative multi-pendulum run. The FFT panel reports simulated peaks near 1.467 Hz, 2.600 Hz and 3.867 Hz and experimental peaks near 1.435 Hz, 2.557 Hz and 3.960 Hz.

7 Mode analysis

Having established the equations of motion governing the pendulum chain, its characteristic collective oscillations are then investigated. Linearizing these equations about equilibrium allows the normal-mode frequencies and mode shapes to be determined, providing the basis for examining how the system’s physical parameters influence the transport and localization of energy within the chain.

7.1 Linearized modal problem

Let $\theta^{(0)}$ be the equilibrium configuration. Writing $\eta = \theta - \theta^{(0)}$ and expanding V to second order gives

$$M\ddot{\eta} + D\dot{\eta} + K\eta = 0, \quad (27)$$

where

$$M = I_{\text{eq}} I_N, \quad D = \gamma I_N, \quad K_{ij} = \left. \frac{\partial^2 V}{\partial \theta_i \partial \theta_j} \right|_{\theta^{(0)}}. \quad (28)$$

In a real open chain, endpoint magnets experience an unbalanced horizontal repulsion, so this work first solves $\nabla V = 0$ and evaluates the Hessian at the calculated equilibrium.

With the state vector

$$X = \begin{pmatrix} \eta \\ \dot{\eta} \end{pmatrix}, \quad (29)$$

the system becomes

$$\dot{X} = AX, \quad A = \begin{pmatrix} 0 & I \\ -M^{-1}K & -M^{-1}D \end{pmatrix}. \quad (30)$$

The $2N$ eigenvalues satisfy

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0 \iff \det(\lambda^2 \mathbf{M} + \lambda \mathbf{D} + \mathbf{K}) = 0. \quad (31)$$

Under the identical-pendulum, nearest-neighbour and free-end approximation, diagonalizing equation (31) with the discrete standing-wave eigenvectors and selecting the underdamped roots gives the closed-form result

$$\begin{aligned} q_m &= \frac{m\pi}{N}, & \phi_n^{(m)} &= \mathcal{N}_m \cos \left[q_m \left(n - \frac{1}{2} \right) \right], \\ \sigma &= -\frac{\gamma}{2I_{\text{eq}}}, & \omega_{d,m} &= \sqrt{\omega_g^2 + 4\kappa \sin^2 \left(\frac{q_m}{2} \right) - \sigma^2}, \\ \eta_n(t) &= \sum_{m=0}^{N-1} A_m \phi_n^{(m)} e^{\sigma t} \cos(\omega_{d,m} t + \varphi_m), & f_m &= \frac{\omega_{d,m}}{2\pi}. \end{aligned} \quad (32)$$

Here $n = 1, \dots, N$ identifies the pendulum, while $m = 0, \dots, N-1$ identifies the collective mode. The constant $\mathcal{N}_m$ normalizes the mode shape, and

$$\omega_g^2 = \frac{K_g}{I_{\text{eq}}}, \quad \kappa = \frac{K_{\text{mag}}}{I_{\text{eq}}}.$$

The $m = 0$ mode describes uniform, gravity-dominated motion. As m increases, neighbouring pendulums move more strongly relative to one another, increasing the magnetic contribution to the modal frequency. For two pendulums, these limits correspond approximately to the in-phase mode $(1, 1)$ and the out-of-phase mode $(1, -1)$.

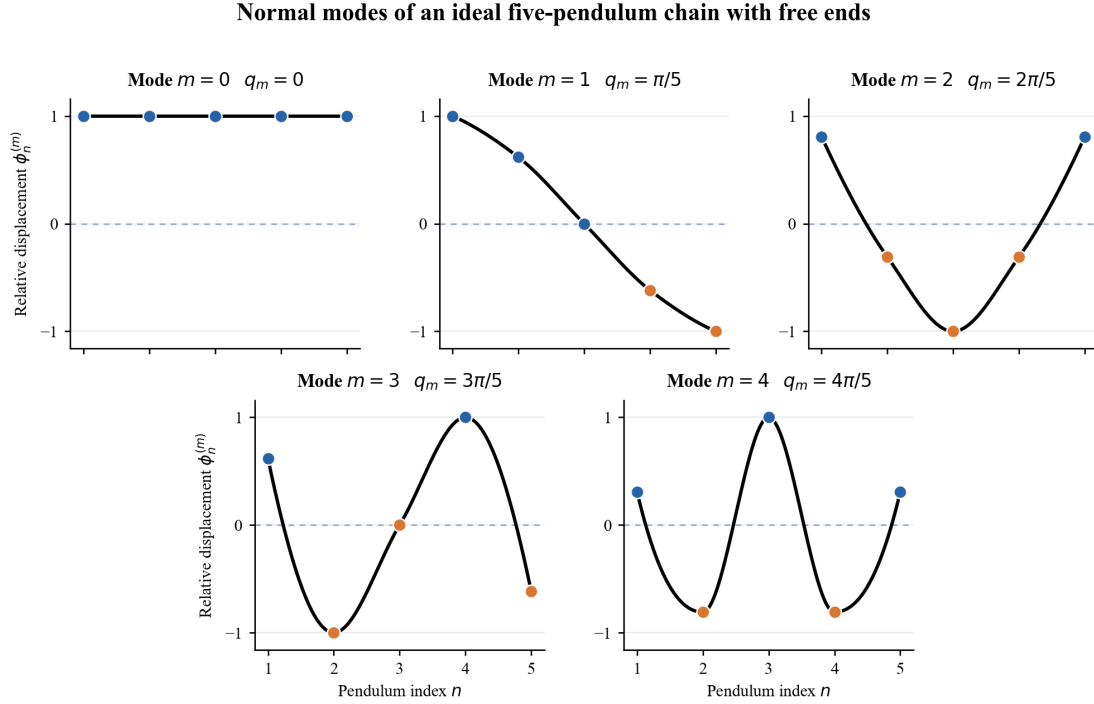


Figure 13: Normal modes of an ideal free-end chain with $N = 5$. Here n is the pendulum index and m is the collective-mode index. The dashed horizontal line marks zero displacement; blue and orange markers indicate positive and negative instantaneous displacement, respectively. The smooth curves are guides to the eye through the discrete mode-shape values. The $m = 0$ mode is uniform and gravity dominated. As m increases, neighbouring pendulums differ more strongly, increasing the magnetic restoring contribution and hence the modal frequency.

7.2 Theory and Experiment Comparison

At a representative release angle of $\theta_0 = 30.3^\circ$, three modes were compared with the model.

Table 2: Representative theory–experiment eigenfrequency comparison.

Mode	Model (Hz)	Experiment (Hz)	Δf (Hz)	$ \varepsilon $ (%)
1	1.425	1.432	+0.007	0.49
2	2.492	2.546	+0.054	2.17
3	3.896	3.952	+0.056	1.44

The mean absolute percentage difference is approximately 1.37% and the frequency RMSE is 0.045 Hz. These differences are comparable to the combined scale of FFT resolution, length/moment uncertainty, and geometric model approximations, so the result supports the measured-parameter model.

The experiments also show the general relative discrepancies of 2.09%, 2.50%, and 6.09% for representative comparisons $N = 3, 4$, and 5 . The increase at $N = 5$ is plausible because magnet heterogeneity, endpoint asymmetry, long-range coupling, and small length errors accumulate as

more oscillators are added.

7.3 Parameter scan trials

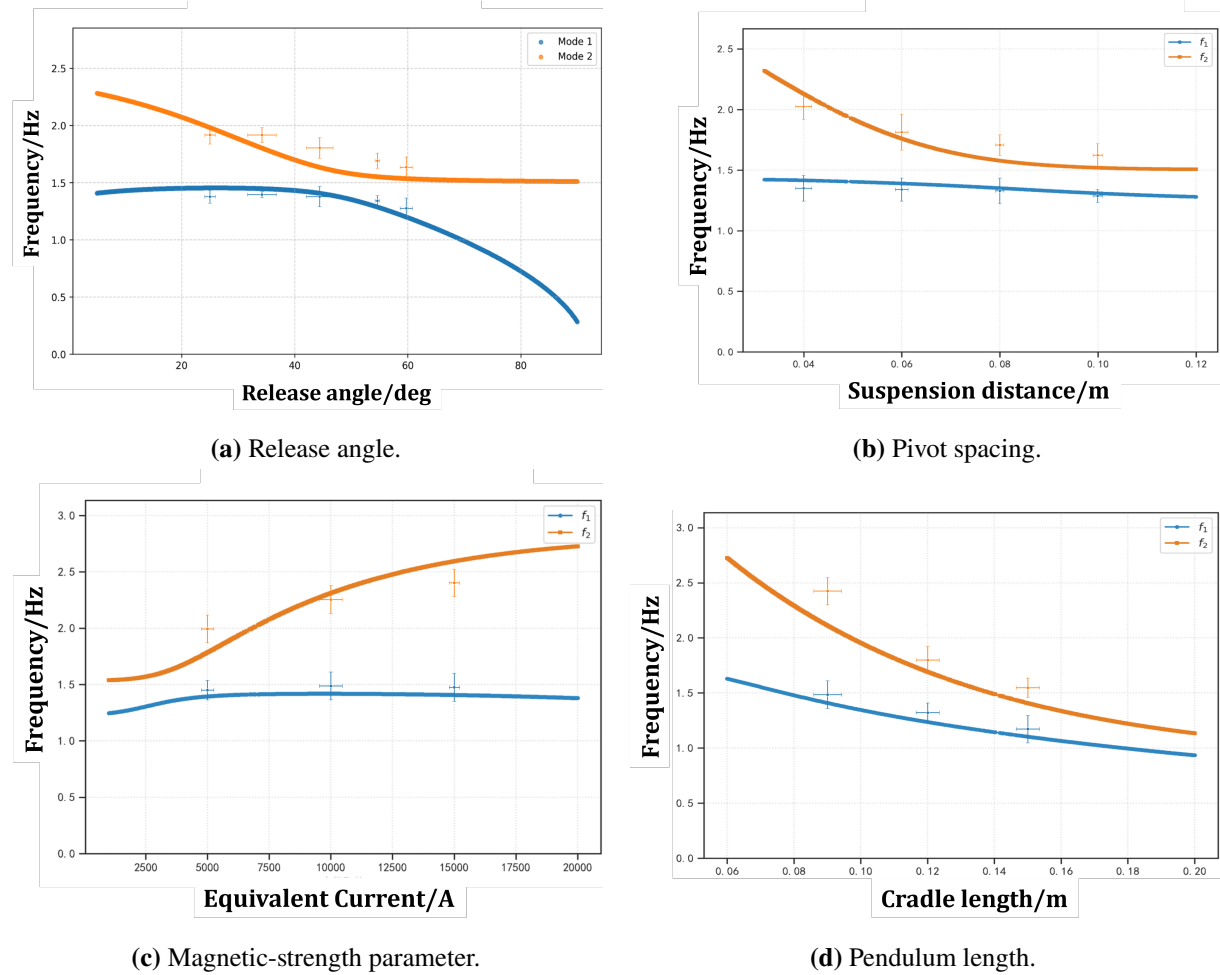


Figure 14: Parameter trials. Smooth curves are theoretical predictions; points are experimental data with error bars.

From the trials, several influences of parameters on the main frequencies can be observed.

- **Release angle:** both branches decrease at sufficiently large amplitude. The gravity contribution softens because $\sin \theta / \theta$ decreases, while the magnetic contribution is sampled over an asymmetric, amplitude-dependent separation.
- **Pivot spacing:** the upper mode decreases strongly as s increases and tends toward the uncoupled gravitational frequency.
- **Magnetic strength:** the upper mode increases markedly with magnetic moment or equivalent current while the lower mode remains near the gravitational branch.

- **Length:** both branches decrease as the pendula lengthen and approach each other as the effective coupling-to-inertia ratio changes. The angular-coordinate gravitational stiffness grows with lever arm, but inertia grows faster, so the natural frequency falls.

8 Chaotic behavior

Modal analysis characterizes the chain's collective oscillations near equilibrium and explains how its dominant frequencies vary with the system parameters. Extending the investigation to larger amplitudes requires considering the full nonlinear dynamics, in which interactions among several modes can produce irregular motion and affect the predictability of energy transfer. It is therefore necessary to determine whether this behavior is complex quasiperiodic beating or genuine deterministic chaos. Two diagnostics were used in the study.

The maximum Lyapunov exponent (MLE) follows the separation of two initially nearby trajectories,

$$\delta(t) \approx \delta_0 e^{\lambda_{\max} t}, \quad \lambda_{\max} = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{\|\delta \mathbf{X}(t)\|}{\|\delta \mathbf{X}(0)\|}. \quad (33)$$

A positive converged MLE indicates exponential sensitivity. Numerically, the Benettin procedure was used: evolve the reference and perturbed states for a renormalization interval τ , record each growth factor $g_j = \|\delta \mathbf{X}_j\|/\delta_0$, renormalize the perturbation, then calculate

$$\lambda_{\max} \approx \frac{1}{K\tau} \sum_{j=1}^K \ln g_j. \quad (34)$$

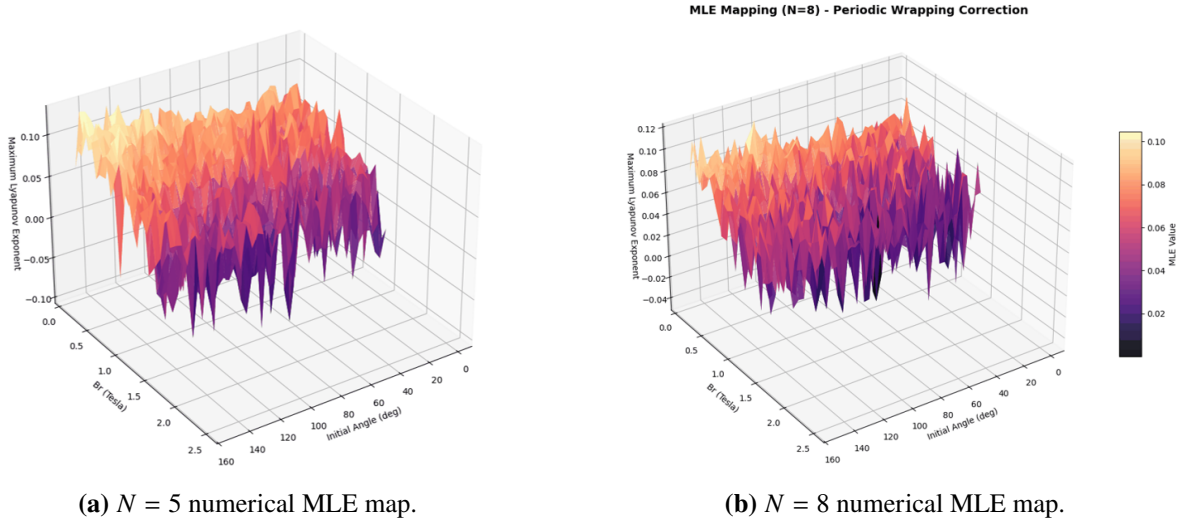


Figure 15: Numerical Lyapunov trials. The simulated maps suggested that large release angle combined with weaker magnetic coupling was more likely to produce a positive MLE, whereas changing N alone did not create a strong influence.

A Poincaré section samples the state whenever a chosen crossing occurs; the parameters are

p_1 at $\theta_3 = 0$ for $N = 5$. A smooth closed curve is consistent with quasiperiodic motion, while an area-filling cloud may indicate chaos.

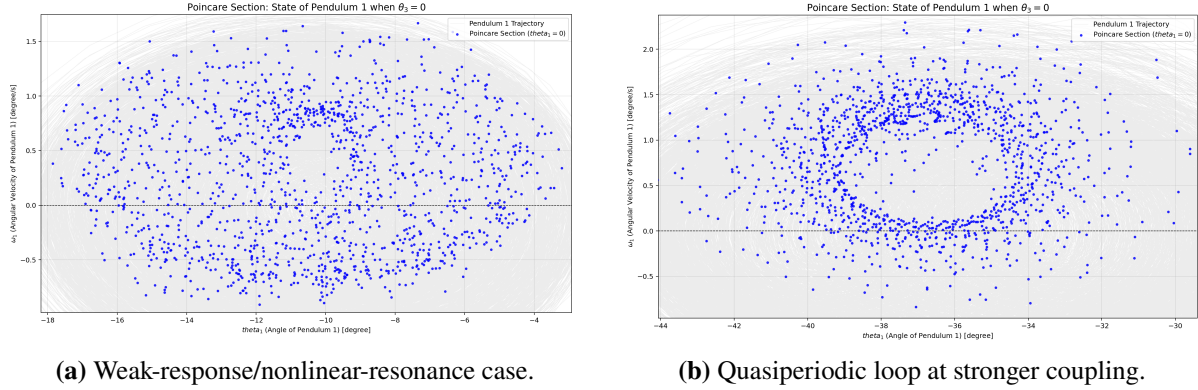


Figure 16: Numerical Poincaré sections from the parameter study. They support the conclusion that much of the investigated range was regular or weakly chaotic rather than globally chaotic.

The MLE maps show that larger release angles combined with weaker magnetic coupling were more likely to produce positive values, while increasing the number of pendulums from $N = 5$ to $N = 8$ did not cause a systematic increase. The Poincaré sections support this result. Stronger coupling produced a well-defined closed loop characteristic of quasiperiodic motion, whereas the weaker-coupling case showed a more dispersed trajectory consistent with irregular or weakly chaotic behavior. Nevertheless, neither diagnostic provided clear evidence of sustained, strongly chaotic motion across the investigated parameter range. The persistence of regular or quasiperiodic dynamics and identifiable spectral peaks indicates that the mode-frequency analysis remains applicable at larger release angles, provided that nonlinear amplitude-dependent frequency shifts are included.

9 Dispersion Relation and Nonlinear Lattice Dynamics

The modal analysis established how the system's discrete frequencies respond to changes in its physical parameters. To understand how these individual modes collectively govern the propagation and localization of energy along the chain, the analysis is now extended to the dispersion relation, which connects oscillation frequency with spatial wave number. The measured magnetic force law provides the coupling strength required for this description and also forms the basis for examining nonlinear lattice effects at larger amplitudes.

9.1 Measured force law

The field measurement was converted into a separation-dependent repulsive force and fitted by

$$F(r) = \beta r^{-\alpha}, \quad \alpha = 2.910, \quad \beta = 1.3253 \times 10^{-4} \text{ N m}^\alpha. \quad (35)$$

For use in the reduced analytical model, the same measurements were also fitted empirically over the restricted interval $x = 3.85$ cm to 12.60 cm:

$$B_{\text{emp}}(x) = 1.6364 \times 10^{-5} x^{-1.910}, \quad (36)$$

where x is expressed in metres and B in tesla. This is a local power-law approximation over the measured interval. The finite-cylinder expression in equation (4) is used for the physical magnet characterization; the power law is used only to obtain a compact reduced interaction model.

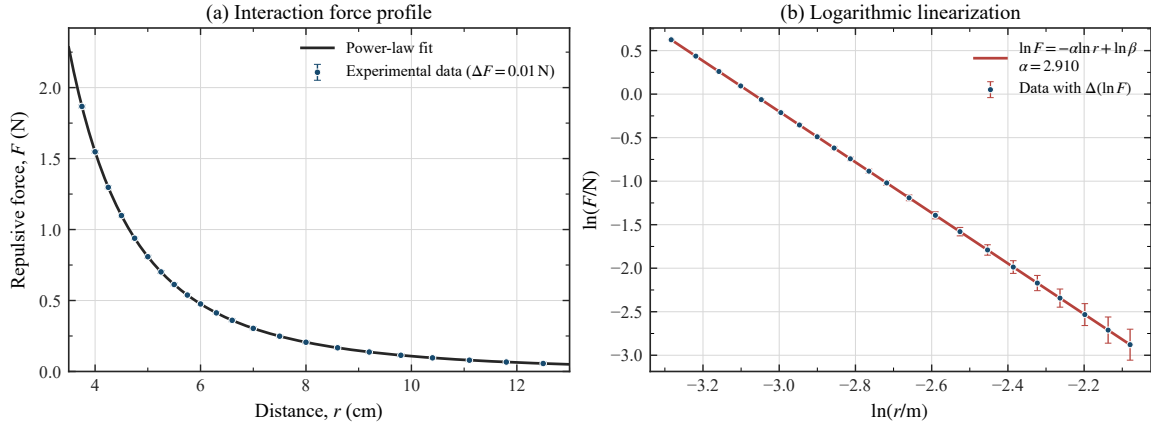


Figure 17: Magnetic field/force power-law characterization used in the reduced dispersion model. The log-log linearization makes the exponent identifiable.

9.2 Derivation of the linear dispersion relation

For small relative angles,

$$r_{n,n+1} \approx s + \ell(\theta_{n+1} - \theta_n). \quad (37)$$

Linearizing $F(r) = \beta r^{-\alpha}$ about $r = s$ gives $|F'(s)| = \alpha \beta s^{-(\alpha+1)}$. Since $\delta r = \ell(\theta_{n+1} - \theta_n)$ and the corresponding torque introduces a second factor of ℓ , the magnetic coupling stiffness is

$$K_m = \alpha \beta \ell^2 s^{-(\alpha+1)}. \quad (38)$$

The gravitational stiffness is $K_g = Mgd$. The interior linearized equation is

$$I_{\text{eq}} \ddot{\theta}_n + \gamma \dot{\theta}_n + K_g \theta_n + K_m (2\theta_n - \theta_{n-1} - \theta_{n+1}) = 0. \quad (39)$$

Ignoring damping when defining the natural branch and substituting a Bloch wave

$$\theta_n = A e^{i(qn - \omega t)}, \quad q = ks, \quad (40)$$

gives

$$\omega^2(q) = \omega_g^2 + 4\kappa \sin^2\left(\frac{q}{2}\right), \quad \omega_g^2 = \frac{K_g}{I_{\text{eq}}}, \quad \kappa = \frac{K_m}{I_{\text{eq}}}. \quad (41)$$

This is an optical-like branch with an on-site gravity gap: $\omega(0) = \omega_g \neq 0$. At the Brillouin-zone edge, $\omega^2(\pi) = \omega_g^2 + 4\kappa$.

The numerical simulation gives $f(0) = 1.444 \text{ Hz}$ and $f(\pi) = 4.144 \text{ Hz}$. These values calibrate the plotted branch. The group velocity is

$$v_g = \frac{d\omega}{dk} = \frac{s\kappa \sin q}{\omega(q)}. \quad (42)$$

It vanishes at $q = 0$ and $q = \pi$, and is largest at an intermediate wavenumber. The curvature $d^2\omega/dk^2$ determines how rapidly a finite-bandwidth wave packet spreads.

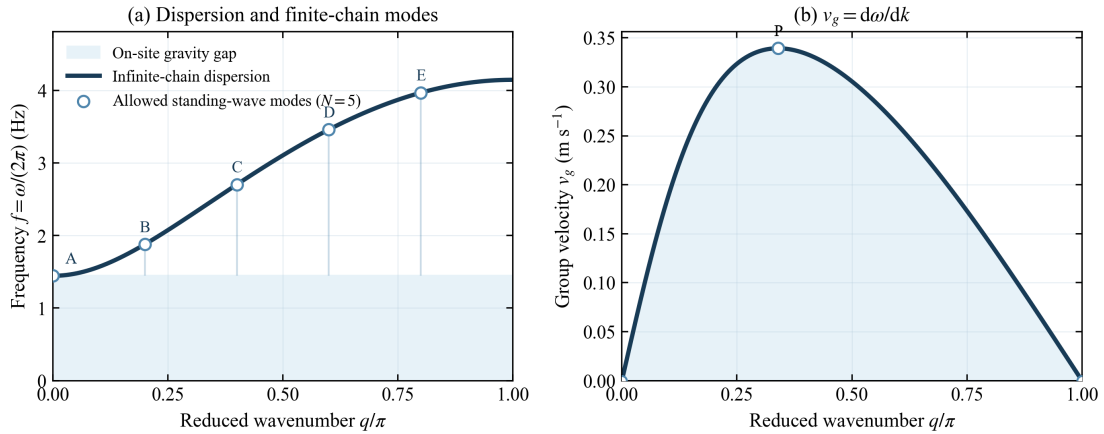


Figure 18: Dispersion relation and group velocity. In (a), the pale-blue region below $f(0)$ represents the gravity-induced stop band. Points A–E denote the theoretical standing-wave modes $m = 0, 1, 2, 3, 4$ of a free-end chain with $N = 5$. In (b), the group velocity is the slope of the dispersion relation, $v_g = d\omega/dk$. It vanishes at both band edges and reaches its maximum at P. The branch is calibrated using $f(0) = 1.444 \text{ Hz}$ and $f(\pi) = 4.144 \text{ Hz}$.

As shown in figure 18(a), gravity keeps the frequency finite at $q = 0$, producing an on-site frequency gap. Magnetic coupling has no effect on the uniform in-phase mode, but its contribution increases with the relative motion between neighbouring pendulums and therefore determines the bandwidth of the dispersion branch. Points A–E connect the continuous dispersion relation to the finite-chain eigenfrequencies studied previously. The normal-mode frequencies and corresponding FFT peaks are discrete samples of the same branch.

Panel (b) provides the corresponding transport interpretation. Because the group velocity is the slope of the dispersion relation, the nearly horizontal band edges have $v_g = 0$, whereas the steeper middle section has the largest group velocity. Consequently, different wavenumber components of a localized disturbance propagate at different speeds, causing a finite-bandwidth wave packet to spread as it travels through the chain.

9.3 Normal-mode analysis of the finite chain

The experimental chain is finite and has free ends. In a nearest-neighbour idealization, the boundary conditions can be written using ghost sites,

$$\theta_0 = \theta_1, \quad \theta_{N+1} = \theta_N. \quad (43)$$

The standing waves are

$$\phi_n^{(m)} = C_m \cos \left[q_m \left(n - \frac{1}{2} \right) \right], \quad q_m = \frac{m\pi}{N}, \quad m = 0, 1, \dots, N-1. \quad (44)$$

Thus the eigenfrequencies measured after release are samples of equation (41) at the allowed q_m . A local end release projects onto many standing waves,

$$\theta_n(t) = \sum_{m=0}^{N-1} Q_m(t) \phi_n^{(m)}, \quad Q_m(0) = \sum_{n=1}^N \theta_n(0) \phi_n^{(m)}. \quad (45)$$

The number and strength of FFT peaks therefore depend both on the spectrum and on the overlap of the initial condition with each mode.

9.4 Long-range correction

Magnetic coupling is not strictly nearest neighbour. If the j th-neighbour stiffness is

$$K_j = \frac{\alpha \beta \ell^2}{(js)^{\alpha+1}}, \quad (46)$$

then

$$\omega^2(q) = \omega_g^2 + \frac{2}{I_{\text{eq}}} \sum_{j=1}^{\infty} K_j [1 - \cos(jq)]. \quad (47)$$

With $\alpha = 2.910$, $K_2/K_1 = 2^{-3.910} \approx 0.066$. This is one likely reason the simplest model degrades as N increases.

9.5 Parameter influence and gravity–magnetic balance

Predicted influence of controllable parameters on the dispersion branch

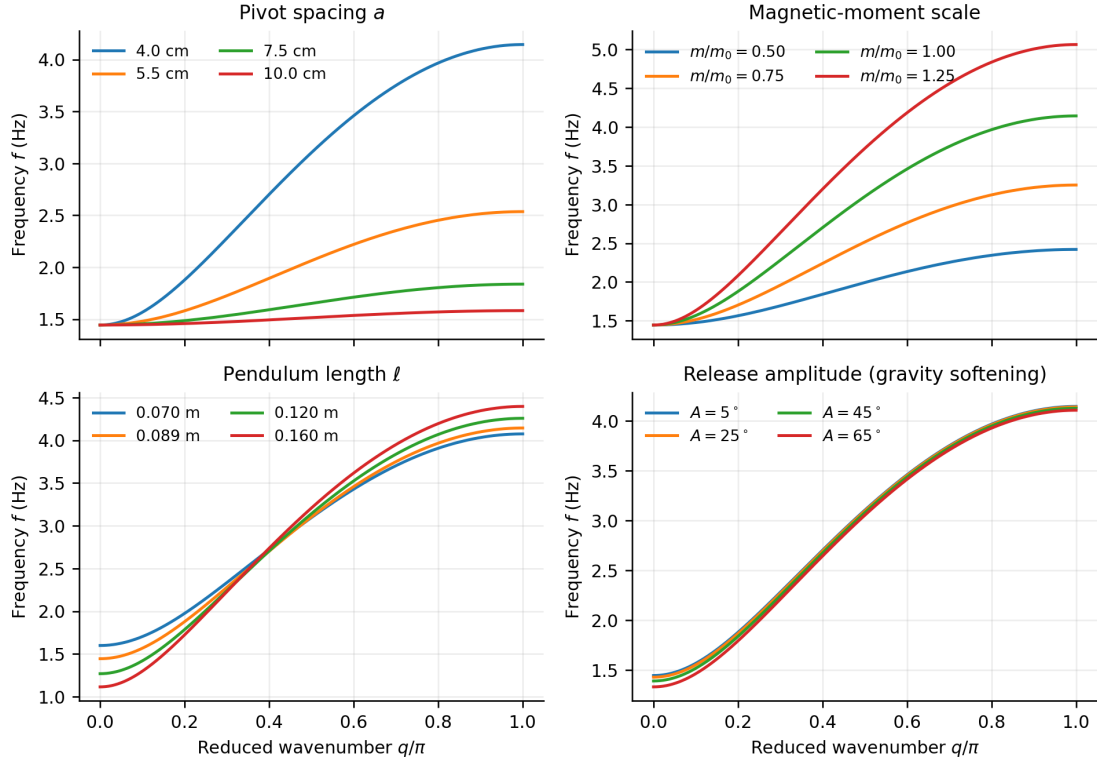


Figure 19: Parameter effects using equation (41), calibrated to $f(0) = 1.444$ Hz and $f(\pi) = 4.144$ Hz. Spacing uses the measured exponent; magnetic moment uses the leading m^2 pair-interaction scaling; the length panel shows leading-order geometry/inertia scaling; release amplitude includes the first-harmonic gravitational softening $2J_1(A)/A$.

The steep scaling

$$K_m \propto \beta \ell^2 s^{-3.910} \quad (48)$$

explains why spacing dominates the upper branch. If the magnetic moment of each identical magnet is scaled by a factor c , dipole-like pair energy and β scale approximately as c^2 .

The fractional contributions to squared frequency are

$$\eta_g(q) = \frac{\omega_g^2}{\omega^2(q)}, \quad \eta_m(q) = \frac{4\kappa \sin^2(q/2)}{\omega^2(q)}, \quad \eta_g + \eta_m = 1. \quad (49)$$

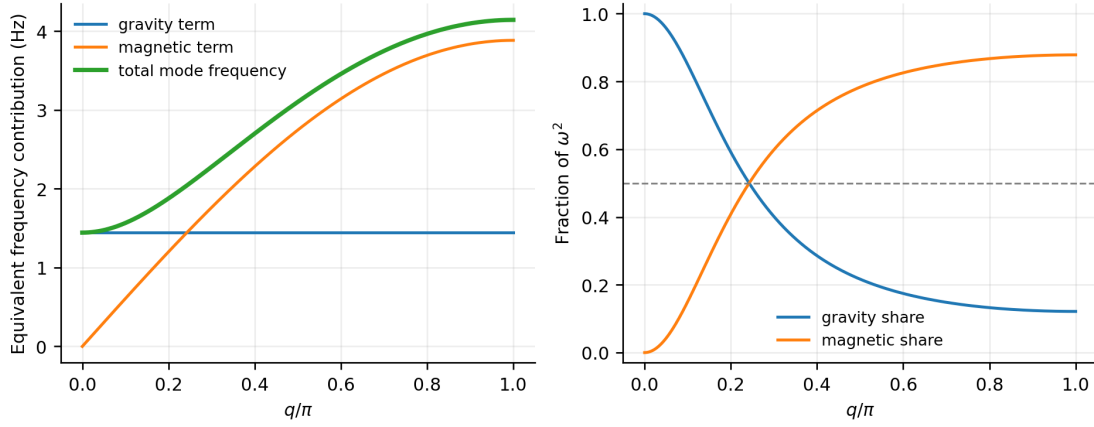


Figure 20: Gravity and magnetic contributions for the calibrated reference branch. At $q = 0$, all pendulums move together and the magnetic share is zero. Toward $q = \pi$, neighbouring pendulums move oppositely and magnetic coupling dominates.

9.6 Nonlinear amplitude-dependent mode evolution

The preceding dispersion relation describes the linear spectrum, although the modal frequencies and energy distribution need not remain fixed when the release amplitude is increased. For an approximately periodic pendulum motion $\theta(t) = \Theta \cos \omega t$, the first harmonic of the gravitational restoring torque gives

$$K_{g,\text{eff}}(\Theta) = K_g \frac{2J_1(\Theta)}{\Theta} = K_g \left(1 - \frac{\Theta^2}{8} + O(\Theta^4) \right). \quad (50)$$

The decrease of $K_{g,\text{eff}}$ with amplitude produces gravitational softening.

The magnetic interaction introduces an additional source of nonlinearity. Using the measured force law $F(r) = \beta_m r^{-\alpha}$ and $r \simeq s + \ell \Delta\theta$, expansion of the pair potential about the equilibrium spacing gives

$$U(s + \ell \Delta\theta) \simeq \frac{K_2}{2} (\Delta\theta)^2 + \frac{K_3}{3} (\Delta\theta)^3 + \frac{K_4}{4} (\Delta\theta)^4, \quad (51)$$

after the constant and equilibrium-balanced linear terms have been removed. The relevant coefficients are

$$K_2 = \alpha \beta_m \ell^2 s^{-(\alpha+1)} = K_m, \quad (52)$$

$$K_4 = \frac{\alpha(\alpha+1)(\alpha+2)}{6} \beta_m \ell^4 s^{-(\alpha+3)}. \quad (53)$$

The quadratic coefficient is therefore the same magnetic stiffness that determines the linear dispersion relation. The higher-order terms cause the modal frequencies to depend on amplitude and allow energy to be exchanged between modes. Depending on the mode shape and equilibrium geometry, the magnetic contribution can either reinforce or oppose the gravitational softening.

Because damping reduces the amplitude during a single record, the instantaneous frequencies

can also drift with time. A short-time Fourier transform separates this drift from a stationary spectrum containing several simultaneous peaks,

$$X_n(t_0, f) = \int x_n(t) w(t - t_0) e^{-i2\pi f t} dt. \quad (54)$$

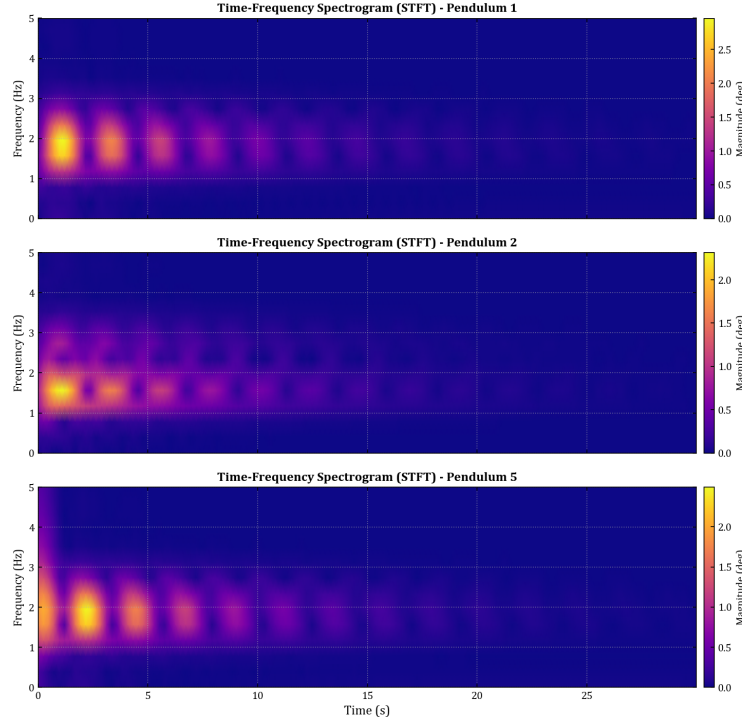


Figure 21: STFT analysis showing how the excited spectral components evolve as the amplitudes decay and energy is redistributed.

The STFT identifies which frequencies are present at each time. To find which modes contain the energy, the displacement is projected onto the linear eigenvectors,

$$Q_m(t) = \sum_{n=1}^N \eta_n(t) \phi_n^{(m)}. \quad (55)$$

The approximate energy and normalized energy fraction of mode m are then

$$E_m(t) = \frac{1}{2} [\dot{Q}_m^2(t) + \omega_m^2 Q_m^2(t)], \quad p_m(t) = \frac{E_m(t)}{\sum_j E_j(t)}. \quad (56)$$

Modal recurrence requires energy to leave the initially excited mode and subsequently return, so a decrease and later recovery of $p_m(t)$ provides more direct evidence than either an FFT or an STFT alone.

To isolate this transfer-and-return mechanism, the magnetic interaction in equation (51) is reduced to an FPUT- β -like model. After scaling the displacement and time, the bond potential is

written as

$$V(\Delta x) = \frac{1}{2}(\Delta x)^2 + \frac{\beta_F}{4}(\Delta x)^4, \quad (57)$$

where the quadratic term represents the same neighbour coupling that produces the linear dispersion, while the quartic term produces nonlinear modal coupling. The gravitational on-site potential, cubic magnetic term, long-range interactions and detailed magnet geometry are not retained in this reduction. Therefore, the model can be used to identify trends in modal recurrence.

Let θ_{ref} be the angular scale used in the nondimensionalization. For an initially excited mode normalized to unit peak displacement,

$$A = \frac{\theta_0}{\theta_{\text{ref}}}, \quad (58)$$

where θ_0 is the maximum initial angular displacement. The combination that controls the quartic nonlinear response is

$$\beta_F A^2 = \frac{K_4}{K_2} \theta_0^2 = \frac{(\alpha + 1)(\alpha + 2)}{6} \left(\frac{\ell \theta_0}{s} \right)^2. \quad (59)$$

Thus increasing the release angle or pendulum length strengthens the effective nonlinearity, whereas increasing the spacing weakens it. The remaining reduced parameters satisfy

$$N = N_{\text{pend}}, \quad \gamma_{\text{red}} = \frac{\gamma}{I_{\text{eq}} \omega_*}, \quad (60)$$

where N is the number of moving pendulums and ω_* is the frequency scale used in the reduced model, and the quantity labelled γ in figure 22 is this dimensionless damping coefficient.

The magnetic moment changes the magnetic coupling relative to gravity according to

$$\frac{K_m}{K_g} \propto \frac{m_{\text{mag}}^2 \ell^2}{M g d s^{\alpha+1}}. \quad (61)$$

The magnetic moment therefore changes the linear bandwidth and modal separations established earlier, while $\beta_F A^2$ describes the additional nonlinear effect of amplitude and geometry.

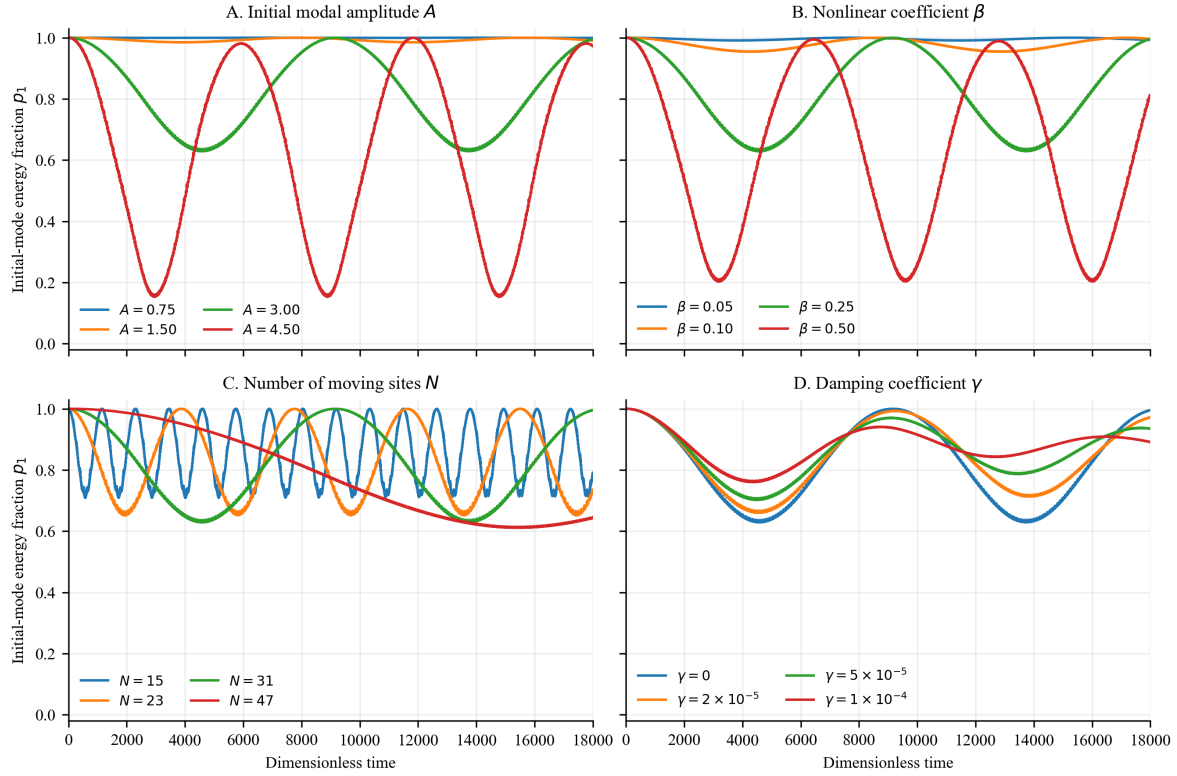


Figure 22: One-at-a-time parameter scans in the reduced FPUT- β model. The plotted quantity p_1 is the fraction of the total modal energy in the initially excited mode. Unless it is being scanned, the parameters are held at $N = 31$, $A = 3.00$, $\beta_F = 0.25$ and $\gamma_{\text{red}} = 0$. For every value of N , all N normal modes are included in the energy normalization, so $p_1(0) = 1$.

Figure 22 shows how the nonlinear modal evolution depends on four distinct controls. In panel A, the two smallest amplitudes keep nearly all the energy in the initial mode. There is therefore little transfer and correspondingly little energy to recur. At $A = 3.00$, p_1 falls to approximately 0.63 before returning close to unity. At $A = 4.50$, it falls to approximately 0.15 and then repeatedly returns, showing that a larger initial amplitude produces deeper and faster nonlinear energy exchange over the displayed range.

Panel B shows the analogous effect of the quartic coefficient. Small β_F produces only weak modulation of p_1 , whereas increasing β_F increases the transfer depth and shortens the recurrence cycle. Panels A and B are consistent with equation (59), since the nonlinear response is controlled mainly by the combined quantity $\beta_F A^2$.

Panel C shows that changing N alters the spacing between the discrete normal modes. The shorter chains complete several rapid returns, while increasing N lengthens the dominant exchange time. For $N = 47$, no complete return occurs within the displayed time interval.

Panel D shows that damping reduces the depth and completeness of the repeated energy exchange. Increasing the reduced damping coefficient decreases the amplitude during the run, which weakens the nonlinear coupling before a full return can occur.

Taken together with the preceding dispersion analysis, these results give a continuous

interpretation of the parameter dependence. The spacing, pendulum length and magnetic moment first determine the linear modal frequencies, bandwidth and mode separation. The release angle and the ratio ℓ/s then set the strength of nonlinear modal coupling, while the number of pendulums controls the density of available modes and damping determines whether a return remains observable. The same experimental parameters that shape the measured mode-frequency spectrum therefore also determine whether finite- amplitude motion remains concentrated in one mode, undergoes coherent modal transfer, or approaches recurrence. The reduced calculation establishes these mechanisms and trends, while recurrence in the experimental apparatus would still require a measured decrease and subsequent return of the complete modal- energy distribution.

9.7 Nonlinear Energy Transport and Localization

The preceding parameter scan showed how amplitude, nonlinear strength, chain length and damping affect modal recurrence once the system is initialized predominantly in one normal mode. Recurrence is, however, only one possible outcome of finite-amplitude motion. A localized release may instead spread through the lattice or remain confined to a small number of sites. The following analysis therefore places modal recurrence within the broader competition between phonon-like spreading, modal recurrence and real-space self-localization.

The preceding STFT identifies which frequencies are present at each time, whereas the present diagnostics determine where the energy is distributed in real space and modal space.

9.7.1 Three dynamical outcomes

For a real-space description, let e_n be the energy assigned to site n and define

$$p_n = \frac{e_n}{\sum_j e_j}, \quad n_E = \sum_n n p_n, \quad \sigma_E^2 = \sum_n p_n (n - n_E)^2, \quad P_x = \frac{1}{\sum_n p_n^2}. \quad (62)$$

Here σ_E measures the spatial width of the excitation and P_x estimates the effective number of occupied sites.

For a modal description, the displacement is projected onto the linear eigenvectors. If Q_m is the coordinate of mode m , its approximate linear modal energy and normalized fraction are

$$E_m = \frac{1}{2} (\dot{Q}_m^2 + \omega_m^2 Q_m^2), \quad p_m = \frac{E_m}{\sum_j E_j}, \quad P_q = \frac{1}{\sum_m p_m^2}. \quad (63)$$

The quantity P_q estimates the effective number of occupied modes. P_x and P_q describe different spaces.

Phonon-like spreading. At sufficiently small release amplitude, the nonlinear frequency shift is weak and the linear dispersion relation remains adequate. A localized release contains several

wave numbers whose different group velocities cause the excitation to broaden. The identifying signatures are increasing σ_E and increasing P_x .

Modal recurrence. Recurrence is tested by initializing one normal mode and following the complete modal-energy distribution. Energy first leaves the initial mode, temporarily occupies a small number of other modes and subsequently returns. The decisive evidence is therefore a decrease and later recovery of the initial-mode fraction, together with P_q returning towards its initial value.

Real-space self-localization. At sufficiently large local amplitude, nonlinear detuning can suppress resonant radiation into the linear phonon band. Energy then remains confined to a small number of neighbouring sites. The corresponding signatures are a persistently small P_x , a large

$$\text{IPR}_x = \sum_n p_n^2 = P_x^{-1}, \quad (64)$$

and a localized oscillation frequency outside, or clearly detached from, the linear band.

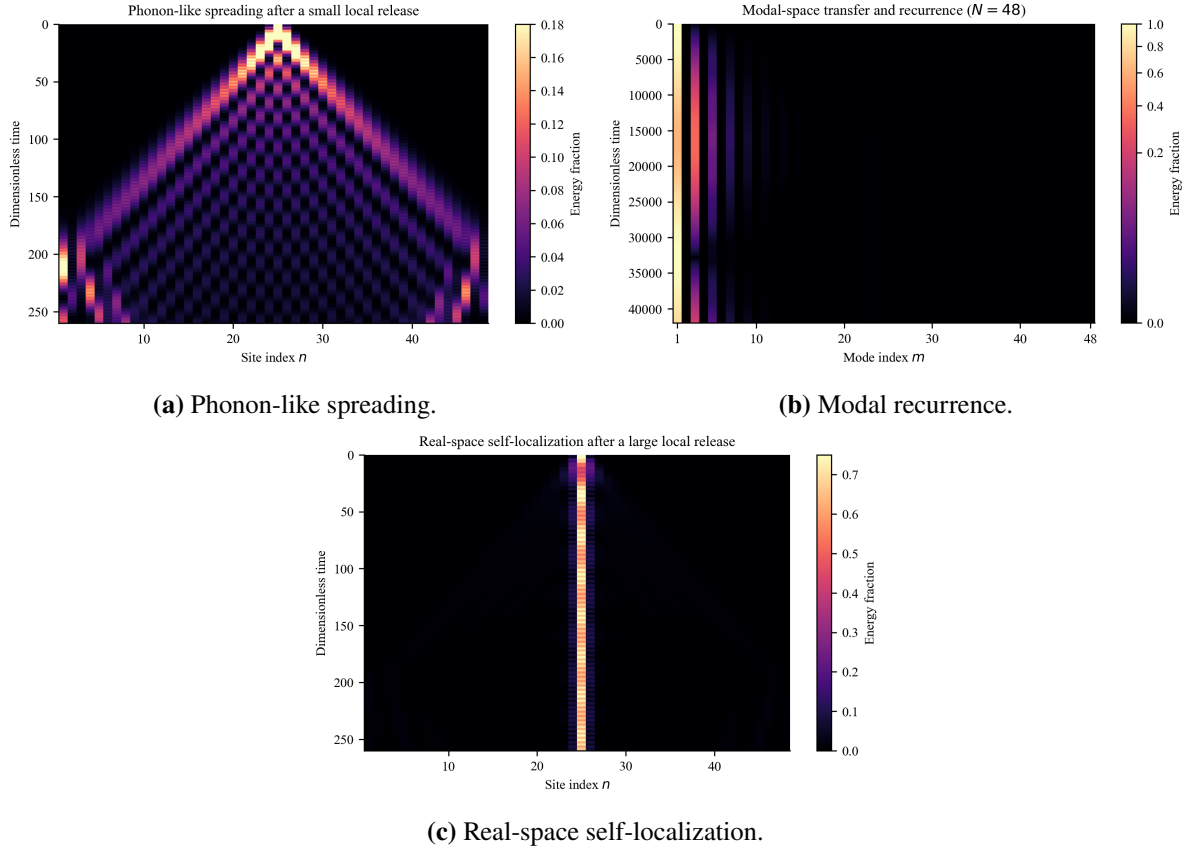


Figure 23: Energy-distribution maps for the three nonlinear-lattice outcomes. Panels (a) and (c) show normalized local-energy fractions in a dimensionless discrete sine–Gordon chain with $N = 48$. In panel (a), a small local release separates into propagating components and occupies an increasing number of sites. In panel (c), a large local release remains concentrated near the initial site. Panel (b) shows all 48 modal-energy fractions of an FPUT- β chain initialized in mode $m = 1$. Energy transfers mainly to low-order odd modes and subsequently returns to the initial mode. The horizontal coordinate therefore represents site number in panels (a) and (c), and mode number in panel (b).

Figure 23(a) contains two spreading fronts. Boundary reflections occur later in the record. In figure 23(b), the periodic transfer away from and back towards $m = 1$ is the defining recurrence signature. In figure 23(c), the narrow vertical band shows that the energy remains spatially concentrated throughout the simulation.

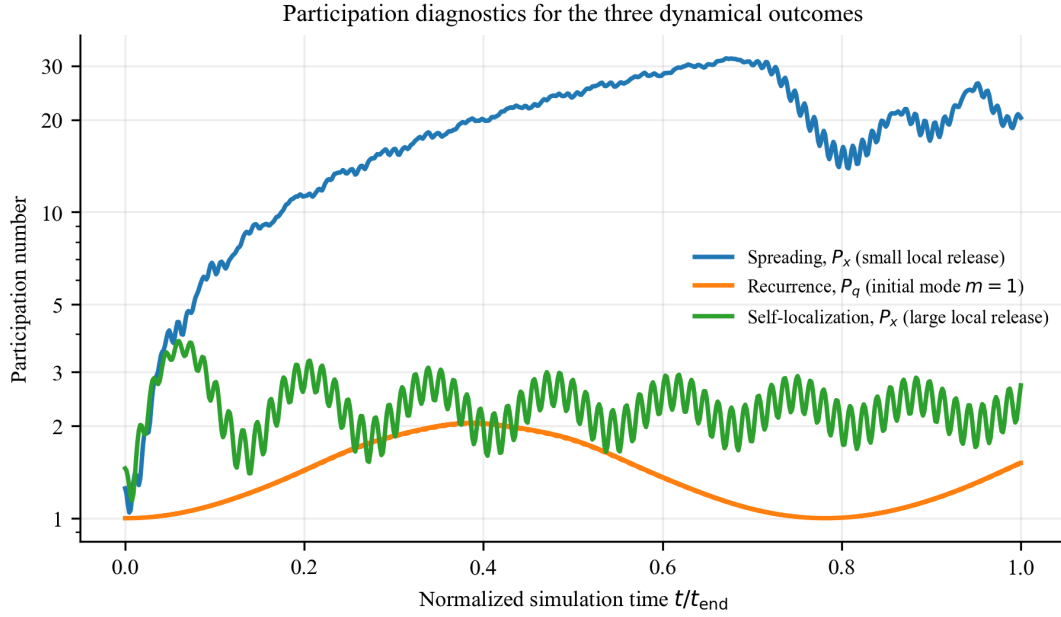


Figure 24: Participation diagnostics for the three dynamical outcomes. The blue and green curves show the real-space participation number P_x , while the orange curve shows the modal participation number P_q . For phonon-like spreading, P_x increases from approximately one site to several tens of sites. During modal recurrence, P_q rises as energy enters additional modes and later returns towards one as energy reconcentrates in the initial mode. For self-localization, P_x remains close to two or three sites. Each time axis is normalized by its own simulation duration.

The participation-number curves provide a compact quantitative distinction between the maps. Spreading produces sustained growth of P_x . Recurrence produces a reversible rise and fall of P_q . Self-localization produces only bounded oscillations of a small P_x .

9.7.2 Estimated boundaries between the three outcomes

The full finite, damped and nonlinear magnetic-pendulum chain does not have closed-form phase boundaries. Approximate boundaries can nevertheless be constructed by comparing the nonlinear frequency shift with the damping linewidth, the discrete modal spacing and the linear phonon bandwidth.

For an isolated pendulum released from angle θ_0 ,

$$\frac{\omega_p(\theta_0)}{\omega_g} = \frac{\pi}{2K(k)}, \quad k = \sin\left(\frac{\theta_0}{2}\right), \quad (65)$$

where $K(k)$ is the complete elliptic integral of the first kind. Define the normalized nonlinear detuning

$$\delta_{\text{nl}} = \frac{|\omega_p(\theta_0) - \omega_g|}{\omega_g}. \quad (66)$$

For the first-mode recurrence calculation, the lower boundary is estimated by

$$\delta_{\text{nl}} \simeq \max \left(\frac{\lambda}{\omega_g}, \frac{\omega_3 - \omega_1}{\omega_g} \right). \quad (67)$$

Below this scale, nonlinear modal coupling is smaller than either the damping linewidth or the separation from the nearest symmetry-allowed mode, and the response is classified as phonon-like spreading.

The upper boundary is estimated from

$$\delta_{\text{nl}} \simeq \frac{\omega_{\text{max}} - \omega_g}{\omega_g}, \quad (68)$$

where $\omega_{\text{max}} - \omega_g$ is the finite-chain phonon bandwidth. Above this scale, the nonlinear frequency is sufficiently detached from the linear band for self-localization to become possible.

The intermediate coloured region is labelled modal recurrence in the figures because it supports resolvable coherent modal transfer. A point is finally identified as recurrent only if its simulated or measured modal-energy distribution leaves and subsequently returns towards its initial state.

The parameter dependence enters mainly through

$$\chi = \frac{K_m}{K_g} \propto \frac{m_{\text{mag}}^2 \ell^2}{M g d s^{\alpha+1}}. \quad (69)$$

Changing s , ℓ or m_{mag} therefore changes the modal spacing and bandwidth and consequently moves both estimated boundaries.

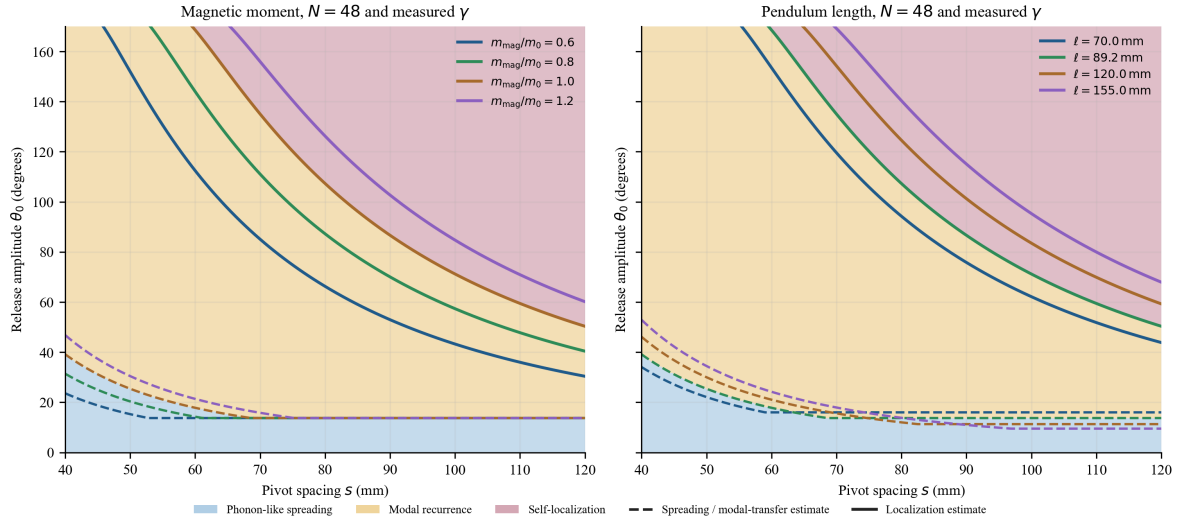


Figure 25: Estimated boundaries in the (s, θ_0) plane with $N = 48$ and the measured damping held fixed. Blue, yellow and red denote phonon-like spreading, modal recurrence and self-localization for the reference configuration. The coloured background applies specifically to $m_{\text{mag}} = m_0$ and $\ell = 89.2$ mm. Dashed curves mark the lower estimate in equation (67), while solid curves mark the localization estimate in equation (68). Panel (a) varies magnetic moment at fixed length. Panel (b) varies pendulum length at fixed magnetic moment. Curves extending above the plotted angular range mean that the corresponding localization condition is not reached below 170° .

Figure 25(a) shows that increasing magnetic moment strengthens the coupling and widens the phonon band. The solid localization boundary consequently moves to larger release angles. Increasing the spacing has the opposite effect because $K_m \propto s^{-(\alpha+1)}$, so the localization threshold falls rapidly with s . Panel (b) shows the corresponding length dependence after the changes in magnetic lever arm and gravitational scale are included. Longer pendulums move the estimated localization boundary upwards over the displayed range.

The dashed curves identify when nonlinear modal transfer first becomes resolvable, whereas the solid curves identify when detuning from the phonon band becomes sufficiently large for spatial localization. The region between them is therefore the parameter interval in which recurrence should be tested using $E_m(t)$ and $P_q(t)$.

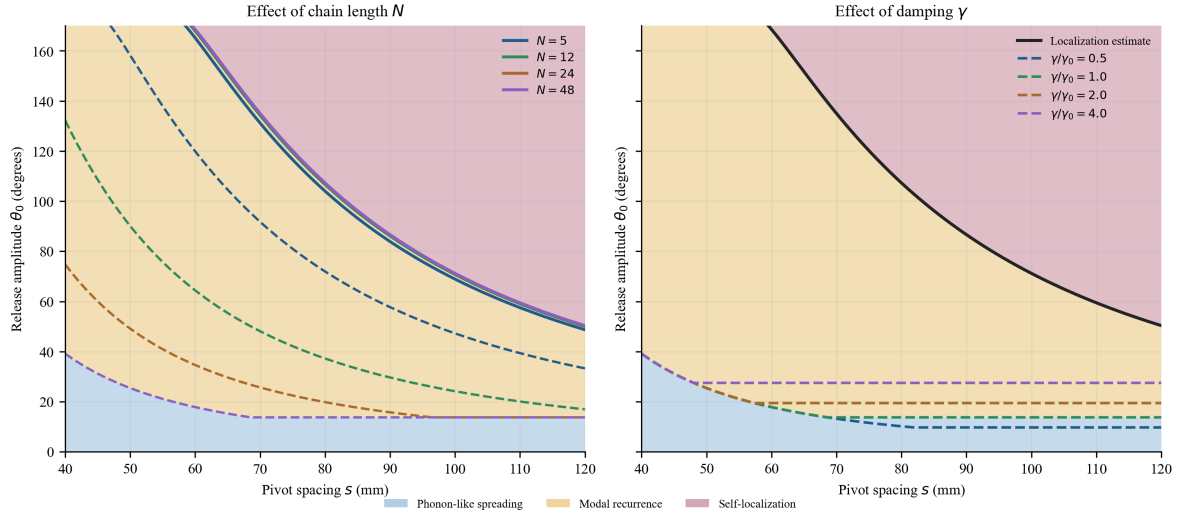


Figure 26: Separate sensitivity of the estimated boundaries to N and γ , with spacing, magnetic moment and pendulum length otherwise evaluated relative to the reference configuration. In panel (a), increasing N reduces the discrete modal spacing and lowers the dashed modal-transfer boundary. The solid localization curves remain close because the overall bandwidth changes much less than the low-mode spacing. In panel (b), increasing γ raises the dashed boundary because nonlinear energy exchange must exceed the damping linewidth to remain observable. The black solid curve is the conservative localization estimate and is independent of damping in this approximation.

The N dependence explains why the five-pendulum calculation did not display the same clear recurrence as the longer reduced chain. A short chain has only a few widely separated modes, so stronger nonlinearity is required before substantial modal transfer develops. Increasing N makes the spectrum denser and lowers this transfer threshold, although it can also lengthen the time required for the modal phases to realign.

Damping affects the observability of the modes. Increasing γ shortens the decay time and can remove energy before a complete modal return occurs. This agrees with panel D of figure 22, where increasing reduced damping makes the recurrence progressively less complete.

Taken together, the recurrence scan and the boundary maps describe two complementary aspects of the dynamics. The FPUT scan shows how energy is transferred and returned once a recurrence-supporting initial condition is chosen. The boundary maps show how the experimental controls move the system between spreading, modal recurrence and self-localization. These boundaries remain model-based estimates until the corresponding spatial and modal-energy diagnostics are evaluated for the measured apparatus.

10 Conclusion

This investigation asked how experimentally relevant parameters affect the normal-mode frequencies of a magnetically coupled pendulum chain and how the resulting spectrum influences energy transport and localization. The results show that the normal-mode spectrum provides a physically

consistent connection between the apparatus parameters and the subsequent dynamics. Using the independently measured magnetic interaction, pendulum geometry, inertia and damping, the Lagrangian model reproduced the measured frequencies to within a few percent. At $\theta_0 = 30.3^\circ$, the measured frequencies were 1.432 Hz, 2.546 Hz and 3.952 Hz, compared with model values of 1.425 Hz, 2.492 Hz and 3.896 Hz. This corresponds to a mean absolute percentage difference of approximately 1.37% and an RMSE of 0.045 Hz. Representative comparisons for $N = 3, 4$ and 5 produced discrepancies of 2.09%, 2.50% and 6.09%, respectively. The increasing discrepancy with N is consistent with the accumulating influence of long-range coupling, endpoint asymmetry, magnet heterogeneity and inertia uncertainty.

The central spectral hypothesis was therefore strongly supported. The nearly in-phase mode remained predominantly controlled by gravity because neighbouring magnets undergo little relative displacement in this mode. Its frequency was correspondingly insensitive to spacing and magnetic moment. Higher modes contain larger neighbour differences and therefore a greater magnetic restoring contribution. Their frequencies increased with magnetic moment and decreased strongly as the spacing increased. Increasing pendulum length reduced the observed frequencies because the increase in rotational inertia outweighed the increase in restoring lever arm. Increasing the release angle produced amplitude-dependent softening, which results from both gravitational and magnetic nonlinearities. Increasing N added allowed standing-wave modes and reduced their separation, while damping had little first-order effect on the natural frequencies but reduced the duration over which modal exchange remained observable.

The dispersion relation explains why these frequency changes matter dynamically. Gravity produces a nonzero lower cutoff, while magnetic coupling controls the bandwidth and the slope of the branch. A local release projects onto several allowed wave numbers, and their different group velocities cause phonon-like spreading. At finite amplitude, nonlinear frequency shifts alter the resonance conditions between the discrete modes. The reduced FPUT calculation showed that increasing the combined nonlinear strength $\beta_F A^2$ deepens and accelerates modal energy exchange, whereas increasing damping makes the subsequent return less complete. The participation diagnostics further distinguished sustained growth of the real-space participation number during spreading, reversible growth and reduction of the modal participation number during recurrence, and a persistently small real-space participation number during self-localization. The numerical Lyapunov and Poincaré analyses showed predominantly regular or quasiperiodic behaviour over the investigated range, with only weaker evidence of irregularity at large release angle and weak coupling. This supports the continued use of a modal description beyond the strictly linear regime, provided that amplitude-dependent frequency shifts are included.

The nonlinear part of the hypothesis is supported at the level of mechanism and numerical trend, but not yet as a complete experimental phase classification. The calculated boundaries show how spacing, magnetic moment, length, N and damping can move the reduced lattice between regions associated with spreading, modal recurrence and self-localization. However, these are

model-based estimates. Experimental recurrence would require the measured modal-energy distribution to leave and subsequently return towards its initial state, while experimental self-localization would require a persistently small P_x together with a localized frequency detached from the measured linear band. The present data therefore establish how the parameters reshape the spectrum and identify the resulting transport mechanisms, but they do not yet constitute an experimental observation of a discrete breather or a complete recurrence boundary.

This distinction clarifies the relationship between the present results and earlier research. Russell *et al.* observed moving breathers in an extended magnetic-pendulum ring after sufficiently strong localized impulses [9]. Their approximately periodic geometry was designed to suppress end reflections and favour travelling localized packets. The present system is instead short, open and damped, so the measured response is dominated by discrete standing modes, reflections and quasiperiodic modal beating. Consequently, the absence of an experimentally confirmed moving breather here is consistent with the results of Russell *et al.*; it identifies the importance of boundary conditions, excitation strength and system size. The present work adds a complementary result by connecting independently measured parameters to the complete finite-chain eigenfrequency spectrum before considering localization.

The comparison with Lorenz *et al.* is similarly complementary [15]. Their two-pendulum system used externally modulated coupling, very small deflections and weak damping to control coherent exchange between two modes. The present passive chain confirms their central physical point that eigenfrequency separation controls energy-transfer timescales, while extending it to several simultaneously available modes, static coupling, finite-amplitude shifts and experimentally varied geometry. In relation to FPUT recurrence and q -breather theory [13], the reduced calculation reproduced the characteristic transfer-and-return mechanism, but the open boundaries and measured damping prevent the conservative, long-duration interpretation from being transferred directly to the experiment. Likewise, the localization mechanism differs from the gap-breather formation observed in diatomic granular chains [3]. Those systems obtain a gap from alternating masses or stiffnesses, whereas the present monatomic chain obtains an on-site frequency gap from gravity and relies on amplitude-dependent detuning from a single optical-like branch.

The research question can therefore be answered with a qualified conclusion. Spacing, magnetic moment, length, release angle, chain size and damping collectively affect energy transport. They first determine the mode frequencies, bandwidth, mode separation and nonlinear detuning, which then control whether energy propagates through several sites, exchanges coherently among a restricted set of modes, or becomes spatially confined. This causal sequence is supported experimentally for the frequency spectrum and parameter trends, and numerically for the subsequent transport and localization mechanisms. A definitive experimental map of recurrence and localization would require longer recordings, controlled modal and local excitation protocols, all-site energy reconstruction and repeated measurements across the predicted boundaries.

11 Evaluation

11.1 Theoretical limitations

Table 3: Evaluation of theoretical and experimental limitations.

Limitation	Influence	Estimated scale	Improvement
Uniform magnetization / fitted $\mathcal{M}$	Field inhomogeneity or a biased axial fit changes magnetic stiffness; high- q modes are affected most.	$u_{\mathcal{M}}/\mathcal{M} \approx 3.6\%$. If $K_m \propto m^2$, its direct relative scale may approach 7.2%, while frequency responds approximately by half the stiffness fraction.	Measure every magnet in the position range actually used; fit $\mathcal{M}, R, H$ with covariance; randomly permute magnets to test heterogeneity.
Nearest-neighbour approximation	Neglects next-nearest and longer-range magnetic coupling, shifting high modes and endpoint equilibria.	$K_2/K_1 \approx 6.6\%$ for $\alpha = 2.910$; frequency effect is mode dependent and smaller than this stiffness ratio.	Use the finite-cylinder pair energy for all $i < j$, then compare AIC/residuals with the nearest-neighbour model.
Linear damping	Real drag depends on speed and Reynolds number; one γ may fit small decay but not large release.	0.1–0.5% frequency effect; γ itself has 2.8% relative uncertainty.	Fit $\gamma_1 \dot{\theta} + \gamma_2 \dot{\theta} \dot{\theta}$ over several amplitudes and test residuals.
Rigid planar motion	Small out-of-plane motion changes true separation and transfers energy to an unmodelled polarization.	0.1–1.2%, potentially larger after imperfect hand release.	Use a mechanical release, dual-camera 3D tracking, stiffer bearings and a plane-constraining guide that does not touch during the run.
Inertia model	Baskets were not included in I_{eq} . A low I predicts frequencies too high.	A 2% inertia bias gives about a 1% frequency bias when one stiffness dominates.	Determine I_{eq} independently from small-angle gravitational period and compare with the CAD/mass calculation.

Limitation	Influence	Estimated scale	Improvement
Small-angle dispersion approximation	At large θ_0 , both the gravitational restoring torque and magnet separation become nonlinear, so the modal frequencies are no longer independent of amplitude.	Between $\theta_0 = 23.2^\circ$ and 49.975° , the measured first frequency changed from 1.4377 ± 0.0035 to 1.3998 ± 0.0320 Hz, a decrease of 2.64%. The second frequency changed from 2.135 ± 0.031 to 2.1273 ± 0.0090 Hz, a decrease of 0.36%; this change is smaller than the combined experimental uncertainty.	Calculate finite-amplitude frequencies using numerical integration, harmonic balance, or Floquet analysis, and compare theory and experiment at the same release angle.

11.2 Measurement and data limitations

Table 4: Evaluation of measurement and data limitations.

Limitation	Influence	Estimated scale	Improvement
Pendulum-length measurement	Uncertainty in ℓ affects the gravitational restoring term and the conversion between measured displacement and angle. Magnetic geometry may introduce additional sensitivity to length.	For $f \propto \ell^{-1/2}$, $\frac{u_f}{f} \approx \frac{1}{2} \frac{u_\ell}{\ell} \approx 0.28\%$, using $u_\ell = 0.0005$ m and $\ell = 0.0892$ m.	Measure the pivot-to-centre-of-mass distance repeatedly with a calibrated instrument, document the measurement reference points, and propagate the length uncertainty through the complete magnetic model.

Limitation	Influence	Estimated scale	Improvement
Finite record duration	A finite observation time limits spectral resolution and may prevent closely spaced modes from being separated. Extending the record does not remove broadening caused by damping.	The FFT-bin spacing is $\Delta f = 1/T$. The reported records give approximately 1–3% relative resolution. A 0.02 Hz difference at 1.4 Hz is approximately 1.4%.	Record more oscillation cycles and estimate frequency using a damped-sinusoid fit, matrix-pencil method, or STFT ridge with bootstrap confidence intervals rather than relying only on the largest FFT peaks.
Video sampling rate	The videos were sampled at approximately 30 fps, corresponding to a time interval of 0.0333 s. This limits the number of measured positions within each oscillation cycle but is distinct from the FFT resolution, which is determined by the total record duration.	For the measured range of approximately 1.4–3.95 Hz, each cycle contains about 21–7.6 frames. The Nyquist frequency is 15 Hz, so aliasing of the measured modes is not expected, although the highest mode is represented by relatively few frames per cycle.	Use the recorded frame timestamps, check for missing frames, and use a higher frame rate for more accurate reconstruction of the highest-frequency mode.

Limitation	Influence	Estimated scale	Improvement
Spatial and tracking resolution	Pixel-level uncertainty in the tracked magnet position propagates into the measured angle and adds noise to the frequency fit.	For equal coordinate uncertainties $u_x = u_y = u_p$, propagation through $\theta = \tan^{-1}(x/y)$ gives $u_\theta = u_p / \sqrt{x^2 + y^2}$. Taking $u_p = 1$ pixel and the measured pivot-to-marker distance of approximately 180–205 pixels gives $u_\theta = 0.0049$ – 0.0056 rad, or approximately 0.28 – 0.32° per tracked point.	Use sub-pixel centroid tracking, increase marker contrast, and repeat the tracking of a selected recording to determine whether the assumed 1 pixel uncertainty is conservative.
Perspective and lens distortion	Camera misalignment rescales the measured horizontal displacement. A constant scale error has little effect on an FFT frequency, but it biases the measured release angle and therefore affects comparisons involving amplitude-dependent frequency.	For a camera misalignment ϕ , the horizontal scale factor is approximately $\cos \phi$. A 5° misalignment gives a $1 - \cos 5^\circ = 0.38\%$ amplitude bias, while a 10° misalignment gives a 1.52% bias.	Align the optical axis normal to the motion plane and place a calibration target in that plane. Apply a planar calibration and lens-distortion correction before converting pixel coordinates into angles.

Limitation	Influence	Estimated scale	Improvement
Release and run-to-run repeatability	Repeated manual releases produce variations in the initial angle, initial velocity, and out-of-plane motion, causing a spread in the extracted frequencies under nominally identical conditions.	The uncertainties obtained from the repeated release-angle trials are 0.0035–0.035 Hz for the first frequency, equivalent to 0.24–2.41%, and 0.0080–0.0585 Hz for the second frequency, equivalent to 0.37–2.69%.	Use a mechanical release and report the mean frequency and repeat-based uncertainty for every configuration.
Damping and non-stationary frequency	If frequency changes as the oscillation amplitude decays, an FFT over the complete record returns an average frequency.	The estimated damping-related frequency effect is 0.1–0.5%, corresponding to approximately 0.001–0.020 Hz over the measured 1.4–3.95 Hz range. This is comparable to the 0.007 Hz discrepancy of the first mode, but is too small by itself to explain the 0.054 Hz and 0.056 Hz discrepancies of the second and third modes.	Fit successive time windows or follow an STFT ridge to test whether the frequency changes during each run.

References

- [1] M. G. Raizen, J. M. Gilligan, J. C. Bergquist, W. M. Itano, and D. J. Wineland. Ionic crystals in a linear paul trap. *Physical Review A*, 45(9):6493–6501, 1992. doi: 10.1103/PhysRevA.45.6493.
- [2] V. J. Sánchez-Morcillo, I. Pérez-Arjona, V. Romero-García, V. Tournat, and V. E. Gusev. Second-harmonic generation for dispersive elastic waves in a discrete granular chain. *Physical Review E*, 88:043203, 2013. doi: 10.1103/PhysRevE.88.043203.
- [3] N. Boechler, G. Theocharis, and C. Daraio. Discrete breathers in one-dimensional diatomic granular crystals. *Physical Review Letters*, 104:244302, 2010. doi: 10.1103/PhysRevLett.104.244302.
- [4] El Mokhtar Hamham, Noé Jiménez, Rubén Picó, Víctor J. Sánchez-Morcillo, Lluís M. García-Raffi, and Kestutis Staliunas. Nonlinear self-collimated sound beams in sonic crystals. *Physical Review B*, 92:054302, 2015. doi: 10.1103/PhysRevB.92.054302.
- [5] Noé Jiménez, Ahmed Mehrem, Rubén Picó, Lluís M. García-Raffi, and Víctor J. Sánchez-Morcillo. Nonlinear propagation and control of acoustic waves in phononic superlattices. *Comptes Rendus Physique*, 17(5):543–554, 2016. doi: 10.1016/j.crhy.2016.02.004.
- [6] Miguel Molerón, Andrea Leonard, and Chiara Daraio. Solitary waves in a chain of repelling magnets. *Journal of Applied Physics*, 115(18):184901, 2014. doi: 10.1063/1.4872252.
- [7] A. Mehrem, N. Jiménez, L. J. Salmerón-Contreras, X. García-Andrés, L. M. García-Raffi, R. Picó, and V. J. Sánchez-Morcillo. Nonlinear dispersive waves in repulsive lattices. *Physical Review E*, 96:012208, 2017. doi: 10.1103/PhysRevE.96.012208.
- [8] Lluís M. García-Raffi, Luis J. Salmerón-Contreras, Noé Jiménez, Mehmed Ahmed, Víctor Sánchez-Morcillo, Rubén Picó, and Juan F. R. Archilla. Nonlinear waves in a chain of magnetically coupled pendula. *Proceedings of Meetings on Acoustics*, 34:045037, 2018. doi: 10.1121/2.0000915.
- [9] F. M. Russell, Y. Zolotaryuk, J. C. Eilbeck, and T. Dauxois. Moving breathers in a chain of magnetic pendulums. *Physical Review B*, 55:6304–6308, 1997. doi: 10.1103/PhysRevB.55.6304.
- [10] Artem Rakcheev and Andreas M. Läuchli. Dynamics of a pair of magnetic dipoles with nonreciprocal interactions due to a moving conductor. *Physical Review B*, 106(17):174435, 2022. doi: 10.1103/PhysRevB.106.174435.
- [11] Sergej Flach and Andrey V. Gorbach. Discrete breathers: Advances in theory and applications. *Physics Reports*, 467(1–3):1–116, 2008. doi: 10.1016/j.physrep.2008.05.002.

- [12] J. F. R. Archilla, Yu. A. Kosevich, N. Jiménez, V. J. Sánchez-Morcillo, and L. M. García-Raffi. Ultradiscrete kinks with supersonic speed in a layered crystal with realistic potentials. *Physical Review E*, 91:022912, 2015. doi: 10.1103/PhysRevE.91.022912.
- [13] S. Flach, M. V. Ivanchenko, and O. I. Kanakov. q-breathers and the fermi–pasta–ulam problem. *Physical Review E*, 73:036618, 2006. doi: 10.1103/PhysRevE.73.036618.
- [14] H. Christodoulidi, C. Efthymiopoulos, and T. Bountis. Energy localization on q-tori, long-term stability, and the interpretation of fermi-pasta-ulam recurrences. *Physical Review E*, 81:016210, 2010. doi: 10.1103/PhysRevE.81.016210.
- [15] Heribert Lorenz, Sigmund Kohler, Anton Parafilo, Mikhail Kiselev, and Stefan Ludwig. Classical analogue to driven quantum bits based on macroscopic pendula. *Scientific Reports*, 13:18386, 2023. doi: 10.1038/s41598-023-45118-y.

A Raw data

This appendix presents the measured raw data. The magnetic-field calibration and run-level frequency records are reproduced first, followed by one complete synchronized three-pendulum time series.

A.1 Data dictionary

Table 5: Definitions of the quantities reported in this appendix.

Field	Unit	Definition
t	s	Elapsed time measured from the first retained video frame.
theta_A, theta_B, theta_C	degree	Raw angular positions of pendulums A, B and C obtained from the synchronized Tracker record. The sign convention follows the image coordinate system.
release_angle_deg	degree	Initial release angle.
f1_hz, f2_hz	Hz	First and second reported dominant FFT frequencies.
B	mT	Measured magnetic-flux density used in the magnetic-field calibration.

A.2 Raw magnetic-field calibration

The first numeric column is not the distance itself. The Logger Pro file stores the transformed coordinate

$$d = x^{-3}, \quad (70)$$

where x is the ruler separation in centimetres. The distance plotted in Figure 4 was therefore reconstructed using

$$x = d^{-1/3}. \quad (71)$$

For example, $d = 0.017523377$ gives $x = 3.85$ cm, while $d = 0.000499906$ gives $x = 12.60$ cm. The values in this appendix and Figure 4 therefore represent the same 26 measurements.

Table 6: Complete magnetic-field calibration data, arranged in two parallel blocks.

Point	$d = x^{-3} \text{ (cm}^{-3}\text{)}$	$x = d^{-1/3} \text{ (cm)}$	$B \text{ (mT)}$	Point	$d = x^{-3} \text{ (cm}^{-3}\text{)}$	$x = d^{-1/3} \text{ (cm)}$	$B \text{ (mT)}$
1	0.017523377	1.75234	8.2130	14	0.003814697	0.38147	2.8090
2	0.015053411	1.50534	7.5910	15	0.003400440	0.34004	2.5700
3	0.013497462	1.34975	7.0150	16	0.002978829	0.29788	2.3150
4	0.012148768	1.21488	6.5100	17	0.002679184	0.26792	2.1230
5	0.010973937	1.09739	6.1300	18	0.002370370	0.23704	1.8890
6	0.010273691	1.02737	5.6110	19	0.002028237	0.20282	1.7020
7	0.009272122	0.92721	5.1890	20	0.001717673	0.17177	1.5670
8	0.008145745	0.81457	4.8760	21	0.001243229	0.12432	1.2400
9	0.007111971	0.71120	4.5650	22	0.001062482	0.10625	1.0659
10	0.006177468	0.61775	4.1420	23	0.000901943	0.09019	0.9950
11	0.005399772	0.53998	3.6440	24	0.000731191	0.07312	0.7871
12	0.004795522	0.47955	3.3880	25	0.000640658	0.06407	0.6807
13	0.004299071	0.42991	3.0590	26	0.000499906	0.04999	0.5399

A.3 Frequency data

Table 7: All run-level release angles and reported dominant frequencies, arranged in two parallel blocks.

Run	$\theta_0 \text{ (}^\circ\text{)}$	$f_1 \text{ (Hz)}$	$f_2 \text{ (Hz)}$	Run	$\theta_0 \text{ (}^\circ\text{)}$	$f_1 \text{ (Hz)}$	$f_2 \text{ (Hz)}$
1-1	31.4	1.460	2.124	4-2	23.1	1.434	2.151
1-2	27.7	1.494	2.241	4-3	24.3	1.438	2.158
1-3	26.4	1.446	2.169	5-1	32.9	1.607	3.080
1-4	34.4	1.447	2.122	5-2	34.1	1.624	n.r.
1-5	35.2	1.429	2.143	5-3	36.2	1.622	n.r.
1-6	34.0	1.439	2.197	6-1	38.8	1.446	1.807
1-7	33.8	1.486	2.193	6-2	37.4	1.442	1.785
2-1	43.7	1.465	2.155	6-3	37.2	1.422	1.761
2-2	43.6	1.484	2.143	7-1	43.4	1.417	2.598
2-3	44.4	1.414	2.159	7-2	44.7	1.437	2.535
3-1	48.1	1.415	2.123	7-3	47.3	1.422	2.586
3-2	48.9	1.479	2.137	8-1	28.7	1.436	2.713
3-3	52.7	1.457	2.119	8-2	26.1	1.500	2.700
3-4	50.2	1.420	2.130	8-3	29.8	1.552	2.716
4-1	22.2	1.441	2.096				

Notes: n.r. means not recorded.

Table 8: Data provenance for the three-mode validation reported in Table 2.

Item	Value or source record
Dataset	3-4cm-0.162m
Configuration	$N = 3$, $s = 0.040$ m, $\ell = 0.162$ m
Release angle	$\theta_0 = 30.3^\circ$
Sampling rate	Approximately 30 frames s^{-1}
Record duration	Approximately 37.7 s
Pendulum A time series	X013S01
Pendulum B time series	X014S01
Pendulum C time series	X015S01
Processed FFT spectra	F013, F014, F015
Experimental modes	1.432 Hz, 2.546 Hz and 3.952 Hz

A.4 Representative raw displacement-time data

Table 9 reports all 962 synchronized time steps from a representative three-pendulum experiment with $s = 0.04$ m and $\ell = 0.155$ m. The record extends from $t = 0$ to $t = 32.033\ 33$ s with a nominal sampling interval of approximately $0.033\ 33$ s.

Table 9: Complete synchronized raw angular time series for the representative three-pendulum experiment, arranged in two parallel blocks.

t (s)	θ_A ($^\circ$)	θ_B ($^\circ$)	θ_C ($^\circ$)	t (s)	θ_A ($^\circ$)	θ_B ($^\circ$)	θ_C ($^\circ$)
0	-18.48246	10.38983	55.70371	16.03333	-26.19322	1.833364	30.42488
0.03333333	-18.48668	10.38981	53.49045	16.06667	-26.67186	2.095921	31.62727
0.06666667	-18.56684	10.38982	47.63483	16.1	-26.51368	2.259969	31.55963
0.1	-18.63933	9.835302	39.23527	16.13333	-25.5181	2.277365	30.3069
0.1333333	-18.79027	7.99778	28.68079	16.16667	-24.23196	2.123985	28.23115
0.1666667	-19.13742	3.559597	20.63538	16.2	-22.91973	1.731858	25.83665
0.2	-20.69726	-3.271026	16.4727	16.23333	-21.9494	1.131443	23.70302
0.2333333	-25.08533	-8.801984	14.38094	16.26667	-21.88243	0.3385023	22.2652
0.2666667	-32.79263	-11.06636	14.00988	16.3	-22.87627	-0.4149617	22.06073
0.3	-41.59768	-11.06279	14.64082	16.33333	-24.88432	-0.9830222	22.70796
0.3333333	-48.86872	-10.50122	16.06532	16.36667	-27.23913	-1.368861	23.86218
0.3666667	-52.12821	-9.815731	18.10478	16.4	-29.52281	-1.526411	25.20938
0.4	-52.12821	-9.340038	20.18789	16.43333	-31.02956	-1.581293	26.19632
0.4333333	-49.50902	-8.797072	22.10768	16.46667	-31.38501	-1.559569	26.64541
0.4666667	-42.32882	-7.958904	23.38965	16.5	-30.69818	-1.509288	26.44037
0.5	-32.63029	-6.415777	23.89389	16.53333	-28.97363	-1.270574	25.68842
0.5333333	-23.50155	-3.58327	23.62728	16.56667	-26.61702	-0.9250382	24.49127
0.5666667	-16.16763	1.508452	23.09853	16.6	-24.26214	-0.3872083	23.44499
0.6	-12.08241	7.416428	23.89179	16.63333	-22.31004	0.3227945	22.75725
0.6333333	-10.56391	11.08081	28.05721	16.66667	-21.36337	1.110032	22.96602
0.6666667	-11.14402	11.44117	35.95404	16.7	-21.55366	1.737003	24.1868
0.7	-13.5189	10.61643	43.82997	16.73333	-22.49833	2.087835	26.14271

Continued on next page

Table 9 continued

t (s)	θ_A (°)	θ_B (°)	θ_C (°)	t (s)	θ_A (°)	θ_B (°)	θ_C (°)
0.7333333	-17.10437	9.579221	49.14368	16.76667	-23.98142	2.248867	28.23106
0.7666667	-21.05742	8.641533	51.12072	16.8	-25.43198	2.272333	30.02749
0.8	-24.99329	7.684724	49.61014	16.83333	-26.55822	2.133221	30.8247
0.8333333	-27.75822	6.347004	45.2391	16.86667	-26.94615	1.929904	30.64585
0.8666667	-29.0717	4.553666	37.09714	16.9	-26.79861	1.667145	29.32409
0.9	-28.98739	1.918989	27.50628	16.93333	-25.96624	1.224331	27.2592
0.9333333	-27.83193	-1.943741	18.5439	16.96667	-24.80111	0.6607516	24.90814
0.9666667	-26.29304	-6.854375	12.25583	17	-23.84413	-0.04199637	22.71396
1	-26.5574	-11.08712	9.041691	17.03333	-23.39982	-0.7601003	21.23676
1.033333	-30.35401	-12.04923	8.636168	17.06667	-23.91047	-1.266721	20.82396
1.066667	-36.08993	-10.66153	10.67958	17.1	-25.21392	-1.488795	21.406
1.1	-41.90921	-9.024335	15.06604	17.13333	-26.93654	-1.504437	22.76891
1.133333	-45.32224	-7.456134	20.64966	17.16667	-28.70792	-1.285948	24.50436
1.166667	-45.32224	-5.985466	26.41633	17.2	-29.77982	-1.037316	26.023
1.2	-43.02879	-4.373367	30.97196	17.23333	-30.05192	-0.7348284	27.10481
1.233333	-36.62199	-2.457504	33.76893	17.26667	-29.28704	-0.397756	27.4286
1.266667	-29.28161	-0.03369316	34.50778	17.3	-27.73718	0.0088049	27.07322
1.3	-21.21355	2.841309	33.39597	17.33333	-25.62113	0.5472232	26.14469
1.333333	-13.82358	6.43264	30.81067	17.36667	-23.44285	1.139061	24.98416
1.366667	-8.866956	10.15104	28.29189	17.4	-21.68853	1.704857	23.90274
1.4	-7.088776	11.88803	28.12173	17.43333	-20.74593	2.092742	23.68758
1.433333	-8.29326	10.90572	31.28269	17.46667	-20.85577	2.313267	24.23293
1.466667	-12.80165	8.765455	36.70674	17.5	-22.04101	2.246746	25.4814
1.5	-19.20145	6.639023	40.36023	17.53333	-23.85048	2.013399	26.9809
1.533333	-26.09389	4.813887	42.17387	17.56667	-25.72553	1.746374	28.35892
1.566667	-32.16788	2.910699	40.60746	17.6	-27.26558	1.448065	28.99253
1.6	-36.50881	0.7999125	37.71214	17.63333	-28.21236	1.090827	28.761
1.633333	-38.52183	-1.507468	31.33258	17.66667	-28.48845	0.6770216	27.64958
1.666667	-38.09581	-3.976288	23.23918	17.7	-27.79994	0.1665195	25.93862
1.7	-35.48005	-6.791349	16.11836	17.73333	-26.62887	-0.3537456	23.92026
1.733333	-31.67181	-9.738142	10.48513	17.76667	-25.38509	-0.8902707	22.05214
1.766667	-28.46557	-11.75039	7.352244	17.8	-24.51021	-1.226438	20.839
1.8	-28.08859	-11.19282	7.384986	17.83333	-24.29042	-1.300924	20.56037
1.833333	-30.30318	-8.483459	10.49053	17.86667	-24.9801	-1.081302	21.2846
1.866667	-33.14588	-5.503826	16.71579	17.9	-26.21615	-0.7438646	22.79957
1.9	-34.9769	-2.904283	24.70853	17.93333	-27.43922	-0.2814469	24.75165
1.933333	-35.1476	-0.7073838	32.6774	17.96667	-28.27806	0.1344185	26.68135
1.966667	-33.37818	1.452508	39.58477	18	-28.58575	0.5628531	28.04932
2	-29.80145	3.526996	43.68215	18.03333	-28.11316	0.9577991	28.67481
2.033333	-24.62634	5.477025	43.68215	18.06667	-26.87638	1.362405	28.42105
2.066667	-18.99541	7.459911	43.68215	18.1	-25.20845	1.597436	27.46166
2.1	-13.70769	9.344908	38.04988	18.13333	-23.37975	1.856822	26.06539
2.133333	-10.09787	10.87391	31.62889	18.16667	-21.90848	2.054736	24.73071
2.166667	-8.80136	10.79598	27.76011	18.2	-21.21074	2.030292	23.93785
2.2	-10.40347	8.169955	26.84297	18.23333	-21.38126	1.765411	23.87673
2.233333	-15.2296	4.47906	28.22895	18.26667	-22.5739	1.474947	24.55686
2.266667	-22.76451	1.337479	29.98789	18.3	-24.39969	1.011628	25.63943
2.3	-31.28148	-1.137773	30.87179	18.33333	-26.44947	0.6280779	26.7169
2.333333	-38.35252	-3.227433	30.40181	18.36667	-28.25888	0.2459675	27.35161
2.366667	-44.25464	-4.908406	28.34288	18.4	-29.25772	-0.1283833	27.39264
2.4	-45.65065	-6.446018	25.0065	18.43333	-29.40397	-0.5099102	26.74054
2.433333	-45.65065	-7.878981	20.86487	18.46667	-28.73861	-0.7649293	25.50134

Continued on next page

Table 9 continued

t (s)	θ_A (°)	θ_B (°)	θ_C (°)	t (s)	θ_A (°)	θ_B (°)	θ_C (°)
2.466667	-41.66036	-9.224549	16.71423	18.5	-27.41582	-0.9831499	23.94691
2.5	-36.44832	-10.35152	13.20582	18.53333	-25.91452	-1.09083	22.45315
2.533333	-30.16377	-10.51004	11.28087	18.56667	-24.58166	-1.01034	21.48069
2.566667	-25.8254	-8.642066	11.53715	18.6	-23.78789	-0.7233105	21.28743
2.6	-24.46847	-4.592732	14.2667	18.63333	-23.85711	-0.2536969	21.95693
2.633333	-24.95834	-0.3365945	20.17921	18.66667	-24.57296	0.2808834	23.47405
2.666667	-25.90301	2.681186	28.45506	18.7	-25.52471	0.7386374	25.34729
2.7	-26.00332	4.801348	37.15662	18.73333	-26.41307	1.068228	27.30262
2.733333	-25.48525	6.234575	44.06025	18.76667	-26.99312	1.316988	28.71365
2.766667	-24.0155	7.266718	48.21087	18.8	-26.94456	1.532242	29.25651
2.8	-21.91168	8.116246	48.21087	18.83333	-26.26603	1.715055	28.99872
2.833333	-19.31522	8.786659	47.02045	18.86667	-25.18712	1.768116	27.88659
2.866667	-16.81853	9.263558	41.49771	18.9	-23.8171	1.768618	26.35867
2.9	-15.07116	9.238957	33.63073	18.93333	-22.70724	1.594084	24.71985
2.933333	-14.45509	7.776408	26.78433	18.96667	-22.06155	1.263899	23.47089
2.966667	-15.54939	4.209593	22.43375	19	-22.26229	0.7450199	22.99048
3	-18.98137	-0.5884893	21.01488	19.03333	-23.31565	0.328408	23.15968
3.033333	-25.35816	-4.4048	21.12111	19.06667	-24.93609	-0.1157999	23.95522
3.066667	-33.32246	-6.432547	21.73756	19.1	-26.78345	-0.4051466	25.04061
3.1	-40.65918	-7.519376	22.24109	19.13333	-28.34877	-0.6439147	25.99007
3.133333	-46.0331	-8.063492	22.29946	19.16667	-29.31667	-0.7671427	26.38334
3.166667	-48.3256	-8.33394	21.93903	19.2	-29.44806	-0.7990005	26.29568
3.2	-48.3256	-8.487249	21.24628	19.23333	-28.67714	-0.7999956	25.63226
3.233333	-43.55204	-8.457681	20.27701	19.26667	-27.32586	-0.7364747	24.66527
3.266667	-37.4561	-8.189543	19.33013	19.3	-25.61678	-0.5748072	23.58144
3.3	-29.43595	-7.191272	18.49952	19.33333	-24.0976	-0.3460129	22.67722
3.333333	-22.66897	-4.63598	18.53602	19.36667	-23.07576	0.07693989	22.41063
3.366667	-18.60607	-0.09830166	19.65922	19.4	-22.82104	0.5977679	22.93606
3.4	-17.20274	4.510135	23.2544	19.43333	-23.33356	0.9926411	24.11694
3.433333	-17.26562	7.350666	29.77869	19.46667	-24.2602	1.26781	25.76609
3.466667	-18.22322	8.197673	38.2731	19.5	-25.35068	1.484385	27.38675
3.5	-19.46279	8.178823	43.63781	19.53333	-26.2417	1.509314	28.53555
3.533333	-20.82333	8.090033	47.42865	19.56667	-26.65904	1.519234	29.01255
3.566667	-21.9451	7.66193	47.42865	19.6	-26.44921	1.480083	28.6204
3.6	-22.74183	7.204649	46.22332	19.63333	-25.93248	1.276048	27.43382
3.633333	-22.92453	6.650216	40.718	19.66667	-25.06245	1.004544	25.86467
3.666667	-22.74522	5.622137	32.64607	19.7	-24.09338	0.6827588	24.21444
3.7	-22.25506	3.385172	24.75973	19.73333	-23.43848	0.2235648	22.81578
3.733333	-21.93591	-0.3552412	18.61389	19.76667	-23.40239	-0.2533998	22.12812
3.766667	-22.98119	-4.960051	15.26066	19.8	-24.0933	-0.5513092	22.2674
3.8	-26.46428	-8.150779	14.37207	19.83333	-25.31129	-0.7498583	23.03875
3.833333	-32.46312	-9.061768	15.03995	19.86667	-26.87054	-0.7340981	24.17764
3.866667	-38.96739	-8.623514	16.94504	19.9	-28.14648	-0.7229545	25.27446
3.9	-43.89785	-7.831722	19.59193	19.93333	-28.97965	-0.6408803	26.20168
3.933333	-46.06794	-7.093946	22.35508	19.96667	-28.92575	-0.4938527	26.4964
3.966667	-46.06794	-6.274334	24.78195	20	-28.1144	-0.373096	26.34706
4	-41.22559	-5.359763	26.44103	20.03333	-26.81211	-0.1559155	25.64629
4.033333	-34.42199	-4.236142	27.00981	20.06667	-25.17442	0.1227835	24.77452
4.066667	-27.02777	-2.442016	26.63364	20.1	-23.67388	0.4560665	23.88435
4.1	-19.99078	0.4096813	25.53011	20.13333	-22.58654	0.822801	23.45536
4.133333	-14.82974	4.251076	24.57165	20.16667	-22.28247	1.178719	23.6315
4.166667	-12.44612	7.843409	25.32675	20.2	-22.7064	1.437588	24.38354

Continued on next page

Table 9 continued

t (s)	θ_A (°)	θ_B (°)	θ_C (°)	t (s)	θ_A (°)	θ_B (°)	θ_C (°)
4.2	-12.36866	9.281076	28.92462	20.23333	-23.71209	1.461925	25.65852
4.233333	-14.25961	8.787497	34.24641	20.26667	-25.00614	1.464647	26.89594
4.266667	-17.73125	7.583163	40.39701	20.3	-26.14393	1.288956	27.81905
4.3	-21.75925	6.495467	43.47929	20.33333	-26.96306	1.088784	28.14176
4.333333	-25.76608	5.403031	43.47929	20.36667	-27.25746	0.9567955	27.82276
4.366667	-28.78697	4.263482	41.3776	20.4	-26.9682	0.6881788	26.79249
4.4	-30.42512	2.917296	34.97004	20.43333	-26.23102	0.3904342	25.32787
4.433333	-30.55676	1.122068	28.85674	20.46667	-25.30242	0.04809198	23.83214
4.466667	-29.41106	-1.233024	21.85815	20.5	-24.44472	-0.2778512	22.51563
4.5	-27.47796	-4.454027	15.78631	20.53333	-24.20016	-0.4677496	21.90223
4.533333	-25.9418	-7.743561	12.00785	20.56667	-24.44284	-0.5778324	21.94241
4.566667	-26.69222	-9.317387	10.84195	20.6	-25.33867	-0.5879658	22.71747
4.6	-29.90327	-8.664144	11.99919	20.63333	-26.5183	-0.4604879	23.90702
4.633333	-34.08588	-7.054094	15.36016	20.66667	-27.50213	-0.2238261	25.28013
4.666667	-37.44604	-5.406456	20.32659	20.7	-28.10203	0.01071552	26.36907
4.7	-38.66694	-3.994233	25.59111	20.73333	-28.10623	0.2673956	27.06731
4.733333	-37.73421	-2.696352	30.24388	20.76667	-27.50521	0.5483045	27.11113
4.766667	-34.74614	-1.228939	33.29247	20.8	-26.37949	0.8631772	26.59822
4.8	-29.74712	0.4176503	34.37661	20.83333	-25.01339	1.102821	25.75075
4.833333	-23.74589	2.428387	33.60164	20.86667	-23.69847	1.399409	24.76779
4.866667	-17.87468	4.777805	31.22976	20.9	-22.83318	1.539169	24.09781
4.9	-13.30652	7.351089	28.23568	20.93333	-22.57287	1.534774	23.94561
4.933333	-10.90848	8.936364	26.44072	20.96667	-22.96684	1.507605	24.37755
4.966667	-11.13164	8.543086	27.06237	21	-24.05618	1.252115	25.11429
5	-14.04774	6.636135	29.6928	21.03333	-25.39372	1.077981	26.08462
5.033333	-19.16739	4.708492	32.56365	21.06667	-26.78336	0.7889268	26.88914
5.066667	-25.15044	3.057706	34.44396	21.1	-27.70873	0.527833	27.24169
5.1	-30.80877	1.655485	35.97007	21.13333	-28.11389	0.262041	27.04215
5.133333	-35.12427	0.2069712	33.84864	21.16667	-27.94536	-0.06708831	26.32165
5.166667	-37.3186	-1.383134	30.4954	21.2	-27.26215	-0.3154417	25.18321
5.2	-37.2883	-3.142855	24.80221	21.23333	-26.16941	-0.5209305	23.96825
5.233333	-35.28467	-5.093613	19.64828	21.26667	-25.12949	-0.6068284	22.93815
5.266667	-31.83199	-7.121352	15.08174	21.3	-24.5053	-0.6567009	22.39156
5.3	-28.17766	-8.518882	12.07772	21.33333	-24.33416	-0.534227	22.45701
5.333333	-26.34259	-8.240876	11.1136	21.36667	-24.85911	-0.2679921	23.21643
5.366667	-26.73622	-6.14396	12.80743	21.4	-25.6291	0.04937626	24.4209
5.4	-28.6176	-3.554329	17.18763	21.43333	-26.36013	0.348719	25.823
5.433333	-30.40266	-1.356647	23.30228	21.46667	-27.03692	0.559981	27.00783
5.466667	-31.17776	0.3853084	30.19403	21.5	-27.22142	0.8078874	27.76059
5.5	-30.73133	1.87815	36.93232	21.53333	-26.96066	0.9564331	27.79908
5.533333	-28.76258	3.288894	40.64224	21.56667	-26.25679	1.065877	27.29164
5.566667	-25.7265	4.604506	40.64224	21.6	-25.12101	1.037234	26.37934
5.6	-21.78569	6.008659	40.64224	21.63333	-24.12224	1.022444	25.23721
5.633333	-17.94348	7.354663	36.06195	21.66667	-23.32822	0.9819502	24.29723
5.666667	-14.71456	8.2716	30.49626	21.7	-23.02598	0.7296363	23.78961
5.7	-12.9506	8.028654	26.49839	21.73333	-23.36586	0.5041204	23.84745
5.733333	-13.39289	6.081881	24.6032	21.76667	-24.31482	0.2366883	24.39143
5.766667	-16.51327	3.18543	24.84776	21.8	-25.62475	0.05128171	25.19263
5.8	-21.90822	0.6042431	25.96575	21.83333	-26.93022	-0.1561417	25.9191
5.833333	-28.51694	-1.189856	27.00376	21.86667	-27.82534	-0.250749	26.45924
5.866667	-34.88376	-2.636227	27.39347	21.9	-28.20346	-0.3560289	26.4706
5.9	-39.66923	-3.832123	26.79155	21.93333	-28.05512	-0.4334027	26.01108

Continued on next page

Table 9 continued

t (s)	θ_A (°)	θ_B (°)	θ_C (°)	t (s)	θ_A (°)	θ_B (°)	θ_C (°)
5.933333	-41.9654	-4.918256	25.22777	21.96667	-27.27616	-0.5066023	25.33688
5.966667	-41.9654	-5.915976	22.76054	22	-26.15683	-0.480802	24.37728
6	-38.87377	-6.836084	19.91931	22.03333	-25.10634	-0.4326598	23.56703
6.033333	-34.59533	-7.411451	17.25334	22.06667	-24.17806	-0.1998738	23.14665
6.066667	-28.65431	-7.159193	15.28006	22.1	-23.84586	0.03439366	23.1413
6.1	-24.50771	-5.568247	14.71546	22.13333	-23.97642	0.3406191	23.70066
6.133333	-22.59146	-2.633221	16.09442	22.16667	-24.58306	0.5698465	24.71257
6.166667	-22.31999	0.566235	20.08923	22.2	-25.35837	0.7806164	26.01866
6.2	-23.16688	2.893927	26.18211	22.23333	-26.1065	0.8414653	27.06434
6.233333	-23.94681	4.382791	32.95284	22.26667	-26.45855	0.9224486	27.67865
6.266667	-24.51104	5.315709	39.41294	22.3	-26.52609	0.96578	27.73048
6.3	-24.43023	6.114267	43.10672	22.33333	-26.13663	0.9613021	27.23059
6.333333	-23.5445	6.695681	43.10672	22.36667	-25.45916	0.9220645	26.29706
6.366667	-22.14091	7.068531	43.10672	22.4	-24.69372	0.751124	25.18178
6.4	-20.46299	7.108908	37.37963	22.43333	-24.04354	0.485931	24.13197
6.433333	-18.88365	6.60942	30.87525	22.46667	-23.85957	0.1820854	23.49327
6.466667	-17.83551	5.088892	25.21367	22.5	-23.99666	-0.0619206	23.33331
6.5	-18.00168	2.404628	21.16194	22.53333	-24.80648	-0.1987672	23.64986
6.533333	-20.13149	-0.9270467	19.50927	22.56667	-25.83319	-0.3408699	24.36284
6.566667	-24.62713	-3.569677	19.53799	22.6	-26.85318	-0.3812444	25.15418
6.6	-30.57046	-4.912781	20.4543	22.63333	-27.5956	-0.4027494	25.84136
6.633333	-36.33295	-5.617828	21.72682	22.66667	-27.91005	-0.4084177	26.13028
6.666667	-40.99802	-5.947881	22.77672	22.7	-27.68366	-0.3834837	26.05459
6.7	-42.61096	-6.06969	23.48353	22.73333	-27.0061	-0.3126721	25.68381
6.733333	-42.61096	-5.978216	23.5404	22.76667	-25.94031	-0.2007706	24.98482
6.766667	-39.44991	-5.688438	23.02432	22.8	-24.85049	-0.1019163	24.19538
6.8	-33.31385	-5.107462	21.91344	22.83333	-23.91878	0.1259617	23.74094
6.833333	-27.30676	-3.947259	20.83232	22.86667	-23.626	0.2114431	23.60371
6.866667	-22.00976	-1.771032	20.13732	22.9	-23.47644	0.4242185	23.76009
6.9	-18.46911	1.264464	20.58154	22.93333	-23.8267	0.5036351	24.31822
6.933333	-17.24888	4.151964	23.02061	22.96667	-24.46824	0.5021142	25.18429
6.966667	-17.71298	5.983961	27.61581	23	-25.36666	0.4628738	26.18161
7	-19.24216	6.57557	33.07185	23.03333	-26.07718	0.4428745	26.80005
7.033333	-21.23638	6.53974	39.32963	23.06667	-26.64089	0.4193899	27.10257
7.066667	-23.23783	6.252667	42.49091	23.1	-26.65918	0.3416367	26.84664
7.1	-24.77154	5.864233	42.49091	23.13333	-26.30927	0.1895614	26.15528
7.133333	-25.52161	5.295195	40.89297	23.16667	-25.70104	0.04247544	25.18857
7.166667	-25.50428	4.432365	36.28583	23.2	-24.97494	-0.1387955	24.15848
7.2	-24.73657	3.192679	28.72539	23.23333	-24.4285	-0.3649864	23.37746
7.233333	-23.71343	1.277738	22.83037	23.26667	-24.18723	-0.4959472	22.99701
7.266667	-22.86887	-1.394232	18.22394	23.3	-24.5027	-0.5760252	23.07011
7.3	-23.50086	-4.327846	15.6211	23.33333	-25.20301	-0.6228006	23.62987
7.333333	-26.00162	-6.11391	15.27082	23.36667	-26.02732	-0.5996911	24.41904
7.366667	-30.29664	-6.612961	16.51945	23.4	-26.71981	-0.5375125	25.31821
7.4	-34.75928	-6.133502	18.94958	23.43333	-27.23084	-0.3482647	26.10413
7.433333	-38.14687	-5.506639	21.95677	23.46667	-27.3025	-0.3164543	26.42561
7.466667	-39.47834	-4.656838	24.76017	23.5	-26.90073	-0.2242058	26.34097
7.5	-38.54576	-3.796887	26.7699	23.53333	-26.05127	-0.05169747	25.96108
7.533333	-35.5208	-2.809523	27.90467	23.56667	-25.155	0.05290793	25.33299
7.566667	-30.57258	-1.692066	27.83871	23.6	-24.11706	0.1818252	24.71868
7.6	-24.88464	-0.1278156	26.8091	23.63333	-23.48309	0.275772	24.20399
7.633333	-19.51489	1.957044	25.41226	23.66667	-23.38503	0.3705932	24.19228

Continued on next page

Table 9 continued

t (s)	θ_A (°)	θ_B (°)	θ_C (°)	t (s)	θ_A (°)	θ_B (°)	θ_C (°)
7.666667	-15.76508	4.388601	24.34626	23.7	-23.62264	0.3880036	24.50867
7.7	-14.30729	6.226426	24.9123	23.73333	-24.40585	0.3870642	25.16575
7.733333	-15.06636	6.721298	27.52264	23.76667	-25.23113	0.3304951	25.8006
7.766667	-17.45956	6.198047	31.25091	23.8	-26.121	0.2496642	26.39272
7.8	-21.05422	5.289245	34.46268	23.83333	-26.75448	0.1334787	26.68648
7.833333	-24.79743	4.23213	37.87554	23.86667	-26.95292	-0.05976757	26.61156
7.866667	-27.99832	3.224626	37.87554	23.9	-26.74452	-0.1658877	26.1141
7.9	-30.25075	2.109041	34.52649	23.93333	-26.18884	-0.3381874	25.31018
7.933333	-31.02966	0.8153994	30.8496	23.96667	-25.41245	-0.4647063	24.44194
7.966667	-30.40171	-0.6827731	25.88968	24	-24.82708	-0.5030338	23.78084
8	-28.7614	-2.480869	20.82442	24.03333	-24.46387	-0.4701539	23.3024
8.033333	-26.675	-4.485921	16.57798	24.06667	-24.46112	-0.4253205	23.42089
8.066667	-25.33204	-6.154961	14.19936	24.1	-24.84617	-0.3514823	23.89404
8.1	-25.86646	-6.599083	13.90849	24.13333	-25.45965	-0.2471604	24.74523
8.133333	-28.2348	-5.736991	15.71544	24.16667	-26.02152	-0.1007842	25.5305
8.166667	-31.08466	-4.301362	19.31155	24.2	-26.51016	0.0639842	26.27007
8.2	-33.29464	-2.963486	23.70437	24.23333	-26.59864	0.1088082	26.60891
8.233333	-33.99901	-1.702317	28.05098	24.26667	-26.41234	0.230546	26.59529
8.266667	-33.20746	-0.6185802	31.35611	24.3	-25.96167	0.2784203	26.21623
8.3	-30.69723	0.488148	33.19801	24.33333	-25.19239	0.3716709	25.5858
8.333333	-26.85682	1.750933	33.17545	24.36667	-24.41509	0.3899534	24.8414
8.366667	-22.46968	3.114189	31.66785	24.4	-23.9479	0.3946188	24.33474
8.4	-18.32999	4.573435	29.15839	24.43333	-23.70391	0.3582927	24.07159
8.433333	-15.26582	5.873573	26.55736	24.46667	-24.08494	0.2918793	24.17196
8.466667	-14.11889	6.229521	25.2344	24.5	-24.73191	0.2188543	24.58063
8.5	-15.17022	5.402116	25.8126	24.53333	-25.56431	0.1069142	25.21295
8.533333	-18.34884	3.745194	27.74287	24.56667	-26.32471	0.02957557	25.76976
8.566667	-22.94495	2.193002	29.84933	24.6	-26.89537	-0.001120667	26.10661
8.6	-27.77744	0.8160729	31.10966	24.63333	-27.13375	-0.05188593	26.14553
8.633333	-31.96607	-0.3571084	31.22238	24.66667	-26.85792	-0.1419293	25.83367
8.666667	-34.68507	-1.467167	29.89276	24.7	-26.35901	-0.1798137	25.35454
8.7	-35.57831	-2.577319	27.23094	24.73333	-25.68962	-0.1678089	24.68732
8.733333	-34.58352	-3.762783	23.7648	24.76667	-25.04938	-0.1603429	24.11348
8.766667	-32.05424	-4.956499	20.08711	24.8	-24.5083	-0.09923913	23.76975
8.8	-28.86633	-5.885854	17.03705	24.83333	-24.36931	-0.007253114	23.8602
8.833333	-25.96706	-6.197963	15.20213	24.86667	-24.49446	0.1762025	24.27719
8.866667	-24.59241	-5.282831	15.21995	24.9	-25.03563	0.198265	24.94791
8.9	-25.0918	-3.4377	17.4004	24.93333	-25.57384	0.2643768	25.67049
8.933333	-26.48805	-1.467535	21.43739	24.96667	-26.07706	0.4061434	26.31735
8.966667	-27.85935	0.08265729	26.45312	25	-26.27972	0.456772	26.64292
9	-28.57754	1.254685	31.31287	25.03333	-26.25078	0.4422487	26.65773
9.033333	-28.25444	2.238911	34.77365	25.06667	-25.90372	0.4265436	26.3181
9.066667	-26.90653	3.124745	37.09714	25.1	-25.29385	0.4164476	25.69954
9.1	-24.67042	3.973438	37.09714	25.13333	-24.78348	0.358256	24.97832
9.133333	-21.9801	4.728421	34.28521	25.16667	-24.33505	0.2661413	24.39252
9.166667	-19.31665	5.337359	30.80909	25.2	-24.2434	0.2676267	24.11209
9.2	-17.25473	5.357016	26.944	25.23333	-24.37432	0.1557127	24.10382
9.233333	-16.48399	4.469317	24.06701	25.26667	-24.8612	0.09326159	24.40646
9.266667	-17.66702	2.550388	23.07368	25.3	-25.54543	0.09906637	24.94954
9.3	-20.80351	0.4817258	23.40274	25.33333	-26.2495	0.0401927	25.50327
9.333333	-25.31105	-1.165141	24.53897	25.36667	-26.69631	0.06860755	25.95327
9.366667	-30.24266	-2.284542	25.67163	25.4	-26.81548	0.08684651	26.06873

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Table 9 continued

t (s)	θ_A (°)	θ_B (°)	θ_C (°)	t (s)	θ_A (°)	θ_B (°)	θ_C (°)
9.4	-34.44813	-3.075373	26.35879	25.43333	-26.68499	0.09907119	25.98647
9.433333	-37.04018	-3.734752	26.21256	25.46667	-26.12018	0.1271494	25.57849
9.466667	-37.66614	-4.317911	25.26018	25.5	-25.53021	0.1738507	25.16019
9.5	-36.14118	-4.791099	23.55923	25.53333	-24.86171	0.2039314	24.64785
9.533333	-33.00491	-5.085243	21.57292	25.56667	-24.49494	0.2254766	24.35799
9.566667	-28.90248	-5.055438	19.64116	25.6	-24.23009	0.3019597	24.38374
9.6	-24.92869	-4.241359	18.40603	25.63333	-24.3051	0.4154793	24.65985
9.633333	-22.30538	-2.431747	18.4159	25.66667	-24.72215	0.4617938	25.20089
9.666667	-21.43289	-0.1652118	20.1869	25.7	-25.27327	0.4379168	25.72563
9.7	-21.49293	0.91831	21.81296	25.73333	-25.83316	0.4238896	26.24361
9.733333	-22.37105	2.532248	26.0835	25.76667	-26.09528	0.414356	26.54733
9.766667	-23.5777	3.45475	30.98884	25.8	-26.16143	0.4017389	26.55626
9.8	-24.63274	3.971378	35.95404	25.83333	-26.04369	0.3669802	26.23904
9.833333	-25.19172	4.256955	38.42596	25.86667	-25.64219	0.3131607	25.70373
9.866667	-25.13943	4.427363	38.42596	25.9	-25.1502	0.2484337	25.07337
9.9	-24.45309	4.509788	36.72986	25.93333	-24.73754	0.1418508	24.51686
9.933333	-23.15979	4.303952	32.26673	25.96667	-24.50567	0.07038428	24.17579
9.966667	-21.70672	3.730821	27.84742	26	-24.54291	0.06195383	24.1645
10	-20.68015	2.445691	23.56886	26.03333	-24.8259	0.05444379	24.44748
10.03333	-20.47801	0.4127263	20.5543	26.06667	-25.36221	0.07790248	24.91514
10.06667	-21.80396	-1.746722	19.31895	26.1	-25.83683	0.09004377	25.40902
10.1	-24.96925	-3.395686	19.59574	26.13333	-26.26568	0.129989	25.83368
10.13333	-29.12277	-4.173031	20.80261	26.16667	-26.4438	0.1803979	26.0942
10.16667	-33.20686	-4.441334	22.39624	26.2	-26.20177	0.1913341	26.0836
10.2	-35.98962	-4.482324	23.82418	26.23333	-25.91281	0.2053141	25.81465
10.23333	-37.16032	-4.340123	24.95347	26.26667	-25.4141	0.2145776	25.39951
10.26667	-36.4047	-4.073924	25.39552	26.3	-24.83652	0.2255157	24.99591
10.3	-33.71719	-3.749642	24.98688	26.33333	-24.50249	0.2170905	24.72722
10.33333	-29.80242	-3.096964	24.06908	26.36667	-24.39958	0.185977	24.60539
10.36667	-25.26893	-2.039421	22.87282	26.4	-24.51901	0.1928442	24.70438
10.4	-21.41229	-0.3725056	21.92201	26.43333	-24.7989	0.170936	25.01117
10.43333	-18.97309	1.660044	22.00303	26.46667	-25.24239	0.1508344	25.4583
10.46667	-18.37855	3.370977	23.65078	26.5	-25.81291	0.1223653	25.84477
10.5	-19.23493	4.186268	26.90342	26.53333	-26.14452	0.09997239	26.1315
10.53333	-21.01096	4.323631	30.69466	26.56667	-26.18632	0.08459675	26.1
10.56667	-23.21627	4.085216	33.83562	26.6	-26.18984	0.04615709	25.86617
10.6	-25.19052	3.728599	35.19994	26.63333	-25.86292	0.01494201	25.40122
10.63333	-26.805	3.272423	35.15849	26.66667	-25.41273	0.03906689	24.97322
10.66667	-27.47212	2.688181	33.40605	26.7	-24.98677	0.04781348	24.51504
10.7	-27.21206	1.875235	30.03806	26.73333	-24.70407	0.07756707	24.22276
10.73333	-26.21647	0.7286364	25.73575	26.76667	-24.68677	0.09596267	24.20836
10.76667	-24.84067	-0.8251873	21.61745	26.8	-24.91376	0.1108455	24.46514
10.8	-23.95766	-2.601881	18.4429	26.83333	-25.20086	0.1380166	24.84558
10.83333	-24.19204	-4.129252	16.91087	26.86667	-25.67922	0.1497726	25.27755
10.86667	-25.98434	-4.653794	17.16944	26.9	-25.95022	0.1812911	25.69871
10.9	-28.90753	-4.479258	18.79	26.93333	-26.06837	0.1882913	25.94647
10.93333	-31.80856	-3.976508	21.3307	26.96667	-26.07904	0.2179602	25.98278
10.96667	-33.93297	-3.387421	24.18233	27	-25.8715	0.2092455	25.75294
11	-34.67492	-2.806333	26.56331	27.03333	-25.42297	0.2017254	25.39388
11.03333	-33.72352	-2.144236	28.22983	27.06667	-25.14279	0.1824321	24.98956
11.06667	-31.21424	-1.311133	28.63613	27.1	-24.85527	0.1621437	24.61297
11.1	-27.56658	-0.3046479	28.10191	27.13333	-24.72628	0.1500478	24.50523

Continued on next page

Table 9 continued

t (s)	θ_A (°)	θ_B (°)	θ_C (°)	t (s)	θ_A (°)	θ_B (°)	θ_C (°)
11.13333	-23.47056	0.8815266	26.72552	27.16667	-24.8624	0.133626	24.53854
11.16667	-19.83028	2.344905	25.03878	27.2	-25.15341	0.110183	24.75137
11.2	-17.53907	3.602871	24.03767	27.23333	-25.56857	0.09467672	25.05946
11.23333	-16.94451	4.24059	24.36124	27.26667	-26.01603	0.09686708	25.37874
11.26667	-18.07919	4.041567	26.14106	27.3	-26.31564	0.0993502	25.65019
11.3	-20.67809	3.423863	28.60358	27.33333	-26.33424	0.08521236	25.70811
11.33333	-23.94049	2.691648	30.85885	27.36667	-26.31417	0.08151882	25.60258
11.36667	-27.07757	1.942167	32.01228	27.4	-25.94186	0.09075151	25.3516
11.4	-29.5577	1.193223	31.79311	27.43333	-25.59153	0.08258721	24.95119
11.43333	-30.82456	0.3719702	30.18041	27.46667	-25.19869	0.09955674	24.69736
11.46667	-30.85794	-0.6089079	27.35375	27.5	-25.04621	0.1091858	24.48332
11.5	-29.7552	-1.775367	23.79234	27.53333	-24.94626	0.1285859	24.47579
11.53333	-27.78796	-3.028	20.32846	27.56667	-25.05995	0.1429743	24.69247
11.56667	-25.91906	-4.037133	17.72514	27.6	-25.30205	0.1650983	24.9971
11.6	-24.94246	-4.416051	16.49344	27.63333	-25.57237	0.1695521	25.42024
11.63333	-25.32921	-4.036828	16.99068	27.66667	-25.86542	0.1850335	25.68031
11.66667	-26.81327	-3.110574	19.04228	27.7	-26.00641	0.1626465	25.83942
11.7	-28.71945	-2.070461	22.2901	27.73333	-25.98824	0.1498322	25.88304
11.73333	-30.14162	-1.160658	25.92374	27.76667	-25.80786	0.1565742	25.72812
11.76667	-30.56442	-0.3421118	29.1805	27.8	-25.50355	0.145446	25.38574
11.8	-29.78842	0.4475503	31.31872	27.83333	-25.24417	0.13049	24.998
11.83333	-27.96986	1.299397	32.0194	27.86667	-25.02268	0.1203993	24.7953
11.86667	-25.36271	2.159203	31.27494	27.9	-24.91153	0.1150075	24.59967
11.9	-22.40687	2.953803	29.40479	27.93333	-25.01835	0.09395637	24.54538
11.93333	-19.63318	3.56009	26.97953	27.96667	-25.23775	0.09343633	24.76875
11.96667	-17.82575	3.741994	24.92797	28	-25.53674	0.09666409	25.00888
12	-17.56904	3.276694	24.05096	28.03333	-25.88844	0.09040571	25.32137
12.03333	-19.06954	2.271876	24.47287	28.06667	-26.07376	0.1042144	25.55016
12.06667	-22.0513	1.200658	25.71535	28.1	-26.12533	0.1107517	25.60887
12.1	-25.79781	0.1986549	27.21538	28.13333	-25.97044	0.1177115	25.56994
12.13333	-29.38722	-0.643243	28.1235	28.16667	-25.71566	0.1231503	25.41447
12.16667	-32.17079	-1.450906	28.29617	28.2	-25.40843	0.125956	25.16803
12.2	-33.63324	-2.24059	27.49153	28.23333	-25.07019	0.1288024	24.871
12.23333	-33.51822	-2.976714	25.71656	28.26667	-24.89538	0.1368777	24.70806
12.26667	-31.90679	-3.509081	23.39876	28.3	-24.85892	0.1446734	24.70757
12.3	-29.36697	-3.990425	20.98396	28.33333	-25.0122	0.1467938	24.86551
12.33333	-26.68698	-4.038743	19.06749	28.36667	-25.25929	0.1502836	24.99616
12.36667	-24.53563	-3.582649	18.12825	28.4	-25.49187	0.1435242	25.35092
12.4	-23.72525	-2.453786	18.70508	28.43333	-25.80175	0.1377744	25.57875
12.43333	-24.13615	-1.065864	20.93092	28.46667	-25.96265	0.1345946	25.66159
12.46667	-25.24989	0.1144476	24.33245	28.5	-25.92092	0.1326542	25.633
12.5	-26.47802	1.123548	28.20465	28.53333	-25.75337	0.1325083	25.43663
12.53333	-27.3216	1.900852	31.42464	28.56667	-25.55236	0.1246367	25.28163
12.56667	-27.3366	2.407503	33.46993	28.6	-25.36453	0.1131358	24.93238
12.6	-26.60707	2.87718	33.85152	28.63333	-25.11101	0.0985619	24.72636
12.63333	-25.10842	3.157297	32.78812	28.66667	-25.08341	0.1062194	24.6439
12.66667	-23.20524	3.272705	30.38763	28.7	-25.0806	0.1128932	24.69369
12.7	-21.34135	3.116007	27.2905	28.73333	-25.23959	0.1174121	24.83923
12.73333	-20.05475	2.532556	24.32259	28.76667	-25.44079	0.1191377	25.0576
12.76667	-19.95783	1.491167	22.48449	28.8	-25.63494	0.1237171	25.36127
12.8	-21.22221	0.2015304	21.95767	28.83333	-25.74565	0.1279864	25.53727
12.83333	-24.05897	-1.014692	22.51013	28.86667	-25.8233	0.1262609	25.62044

Continued on next page

Table 9 continued

t (s)	θ_A (°)	θ_B (°)	θ_C (°)	t (s)	θ_A (°)	θ_B (°)	θ_C (°)
12.86667	-27.53439	-1.838974	23.69831	28.9	-25.78455	0.1274035	25.56088
12.9	-30.9118	-2.404787	24.98055	28.93333	-25.57501	0.1308974	25.38509
12.93333	-33.41658	-2.700048	25.81714	28.96667	-25.33623	0.1313299	25.15009
12.96667	-34.60426	-2.928309	25.93755	29	-25.07762	0.1293226	24.99919
13	-34.04683	-2.990865	25.40843	29.03333	-25.04599	0.1310608	24.80824
13.03333	-32.16233	-2.957257	24.31653	29.06667	-25.01485	0.130625	24.87435
13.06667	-29.10612	-2.756871	22.89039	29.1	-25.11392	0.1310969	25.00076
13.1	-25.86627	-2.144574	21.50755	29.13333	-25.30479	0.1301097	25.15509
13.13333	-23.16038	-1.146926	20.8022	29.16667	-25.51841	0.1296252	25.36096
13.16667	-21.53821	0.1266249	21.23183	29.2	-25.66474	0.1275165	25.50363
13.2	-21.41019	1.44639	22.93973	29.23333	-25.75833	0.1269915	25.60577
13.23333	-22.24547	2.385846	25.73756	29.26667	-25.70011	0.1249627	25.57663
13.26667	-23.65286	2.919525	29.04776	29.3	-25.4824	0.1259843	25.41228
13.3	-25.12831	3.045416	31.87995	29.33333	-25.34668	0.1258032	25.28381
13.33333	-26.12516	3.137396	33.46385	29.36667	-25.18866	0.120241	25.14174
13.36667	-26.4683	3.053357	33.65204	29.4	-25.05906	0.1196943	25.00132
13.4	-26.21219	2.849935	32.21262	29.43333	-24.99227	0.1215234	24.98352
13.43333	-25.38944	2.425333	29.53012	29.46667	-24.98768	0.1229596	25.01884
13.46667	-24.14233	1.776423	26.18534	29.5	-25.12428	0.1236464	25.15538
13.5	-22.97727	0.840362	23.04485	29.53333	-25.34061	0.1235505	25.37514
13.53333	-22.59696	-0.3815753	20.81917	29.56667	-25.47348	0.124697	25.49125
13.56667	-23.37694	-1.647398	19.89155	29.6	-25.66587	0.1303513	25.63183
13.6	-25.31721	-2.416939	20.30614	29.63333	-25.642	0.1354669	25.63196
13.63333	-28.07344	-2.66571	21.60687	29.66667	-25.58639	0.1329255	25.55458
13.66667	-30.80268	-2.615714	23.34072	29.7	-25.42118	0.1310337	25.37955
13.7	-32.69949	-2.374892	25.14691	29.73333	-25.27399	0.1305877	25.22176
13.73333	-33.37145	-2.095234	26.36398	29.76667	-25.12211	0.1300496	25.06689
13.76667	-32.6985	-1.757885	26.87424	29.8	-25.02802	0.12893	24.96941
13.8	-30.59589	-1.318258	26.6284	29.83333	-24.99293	0.1297036	24.96331
13.83333	-27.64809	-0.7275154	25.75181	29.86667	-25.12277	0.1295778	25.02106
13.86667	-24.31723	0.07514248	24.4937	29.9	-25.20818	0.1256727	25.17
13.9	-21.57421	1.12885	23.43786	29.93333	-25.33781	0.1270795	25.32172
13.93333	-19.82663	2.267001	23.18825	29.96667	-25.47443	0.1276329	25.45734
13.96667	-19.51176	3.083963	24.08216	30	-25.45103	0.1279491	25.44357
14	-20.47617	3.38254	26.05817	30.03333	-25.50101	0.1279871	25.36072
14.03333	-22.36148	3.271115	28.54995	30.06667	-25.31408	0.1276239	25.23324
14.06667	-24.57485	2.883719	30.68581	30.1	-25.21441	0.1278461	25.10301
14.1	-26.62222	2.478429	31.77488	30.13333	-25.09861	0.1280378	25.02638
14.13333	-27.92739	2.028124	31.76135	30.16667	-24.95166	0.1271785	24.93115
14.16667	-28.46717	1.533977	30.3884	30.2	-24.96224	0.127701	24.98663
14.2	-28.04854	0.9360021	27.81521	30.23333	-25.00688	0.1274508	25.0468
14.23333	-26.90923	0.09649818	24.82616	30.26667	-25.1181	0.127426	25.20328
14.26667	-25.56624	-0.8934855	21.90332	30.3	-25.33742	0.1274314	25.26738
14.3	-24.46557	-1.90848	19.83658	30.33333	-25.34338	0.1274023	25.42364
14.33333	-24.33951	-2.490077	19.01352	30.36667	-25.44393	0.1265732	25.40229
14.36667	-25.3564	-2.690558	19.41828	30.4	-25.41255	0.1269123	25.37623
14.4	-27.22275	-2.267642	21.08635	30.43333	-25.28905	0.1261414	25.28812
14.43333	-29.17789	-1.782088	23.39724	30.46667	-25.1767	0.1264701	25.1759
14.46667	-30.63892	-1.224643	25.8096	30.5	-25.1594	0.1266911	25.11582
14.5	-31.02811	-0.6726229	27.77277	30.53333	-24.9978	0.1266269	24.98508
14.53333	-30.36292	-0.1787566	29.03519	30.56667	-25.04828	0.1266506	25.01737
14.56667	-28.51975	0.4089523	29.31823	30.6	-25.0717	0.1266755	25.04083

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Table 9 continued

t (s)	θ_A (°)	θ_B (°)	θ_C (°)	t (s)	θ_A (°)	θ_B (°)	θ_C (°)
14.6	-26.07985	1.098516	28.3927	30.63333	-25.21482	0.1264938	25.10562
14.63333	-23.25491	1.909906	26.96115	30.66667	-25.24198	0.1265893	25.22781
14.66667	-20.90705	2.566724	25.39523	30.7	-25.42431	0.1258738	25.26745
14.7	-19.47015	3.030941	24.29844	30.73333	-25.4093	0.1263412	25.34225
14.73333	-19.30275	3.082004	24.32055	30.76667	-25.40113	0.1264584	25.40621
14.76667	-20.61136	2.661048	25.39713	30.8	-25.37766	0.1264622	25.36013
14.8	-22.81437	2.086852	26.92008	30.83333	-25.2445	0.1266808	25.24269
14.83333	-25.5276	1.476354	28.39575	30.86667	-25.17188	0.1266829	25.18027
14.86667	-28.06046	0.9323198	29.31448	30.9	-25.1935	0.1265887	25.13953
14.9	-29.85631	0.3866329	29.30789	30.93333	-25.14222	0.126508	25.10151
14.93333	-30.5834	-0.1714653	28.33427	30.96667	-25.11904	0.1258746	25.08008
14.96667	-30.26758	-0.754709	26.47272	31	-25.14885	0.1261862	25.12468
15	-28.96646	-1.405912	24.20152	31.03333	-25.25649	0.1264704	25.20958
15.03333	-27.13308	-2.05308	21.87656	31.06667	-25.26753	0.127263	25.19914
15.06667	-25.44246	-2.395812	20.121	31.1	-25.33432	0.1272347	25.35739
15.1	-24.53672	-2.345645	19.44094	31.13333	-25.36111	0.1271727	25.35335
15.13333	-24.66031	-1.781241	20.0609	31.16667	-25.37404	0.1270977	25.32391
15.16667	-25.57505	-1.006838	21.91217	31.2	-25.27164	0.1270921	25.2508
15.2	-26.8924	-0.2627681	24.57748	31.23333	-25.21854	0.127023	25.20718
15.23333	-28.02963	0.383176	27.45413	31.26667	-25.19598	0.1291148	25.20876
15.26667	-28.41914	0.9340515	29.85443	31.3	-25.21922	0.129224	25.18718
15.3	-28.11448	1.451954	31.09416	31.33333	-25.26991	0.1285156	25.23677
15.33333	-27.08587	1.975026	31.22219	31.36667	-25.30004	0.1283211	25.28632
15.36667	-25.45493	2.438159	30.23836	31.4	-25.41242	0.1287172	25.35796
15.4	-23.43403	2.699077	28.35874	31.43333	-25.46559	0.1286865	25.44004
15.43333	-21.60059	2.726337	26.20057	31.46667	-25.46412	0.1285554	25.41962
15.46667	-20.3633	2.485277	24.56298	31.5	-25.45496	0.128357	25.38992
15.5	-20.34665	1.939098	23.6779	31.53333	-25.4207	0.128309	25.36142
15.53333	-21.62167	1.196319	23.95764	31.56667	-25.36734	0.128309	25.34083
15.56667	-23.93473	0.4899161	24.85343	31.6	-25.3016	0.128309	25.27444
15.6	-26.7421	-0.1325713	25.96771	31.63333	-25.24574	0.128309	25.2072
15.63333	-29.25619	-0.6246212	26.99056	31.66667	-25.23458	0.1283747	25.2495
15.66667	-31.01225	-1.070631	27.32193	31.7	-25.27682	0.1278946	25.27679
15.7	-31.63556	-1.372446	26.99815	31.73333	-25.31333	0.1280885	25.29387
15.73333	-31.15498	-1.669302	26.00907	31.76667	-25.35206	0.1281142	25.33945
15.76667	-29.48379	-1.967958	24.44986	31.8	-25.46994	0.1284232	25.37767
15.8	-27.38781	-2.016481	22.85466	31.83333	-25.43053	0.1280213	25.40672
15.83333	-25.18117	-1.799775	21.52807	31.86667	-25.40439	0.1283382	25.37798
15.86667	-23.49814	-1.225945	20.89584	31.9	-25.38946	0.1282754	25.35929
15.9	-22.87903	-0.4020362	21.41115	31.93333	-25.37611	0.1282484	25.33652
15.93333	-23.21447	0.4207689	23.08493	31.96667	-25.35512	0.128243	25.2953
15.96667	-24.23511	1.018288	25.65455	32	-25.27808	0.1324733	25.26927
16	-25.32562	1.526851	28.30525	32.03333	-25.30848	0.111346	25.23311